

Strategies for integrating disparate social information

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Significance statement

Social information pervades human life, influencing behavior for the better (e.g., adopting valuable ideas) or worse (e.g., spreading false news). Both on- and offline, people routinely interact in groups harboring disparate opinions, views and behaviors. However, we still know little about the strategies people use to integrate disparate social information. Combining experimental and cognitive modelling approaches, we show that the impact of social information strongly depends on its distribution: agreement among peers increases people's tendency to follow them, and peers confirming one's initial beliefs leads people to ignore others who disagree. Our results contribute to understanding the nature of individuals' social information use, and delineate when groups can build consensus or are bound for deepening polarization.

Abstract

Our increasingly interconnected world provides virtually unlimited opportunities to observe the behavior of others. This affords abundant useful information but also requires navigating complex social environments with people holding disparate or conflicting views. It is, however, still largely unclear how people integrate information from multiple social sources that (dis)agree with them, and among each other. We address this issue in three steps. First, we present a judgment task in which participants could adjust their judgments after observing the judgments of three peers. We experimentally varied the distribution of this social information, systematically manipulating its variance (extent of agreement among peers) and its skewness (peer judgments clustering either near or far from the participant's). As expected, higher variance among peers reduced their impact on behavior. Importantly, observing a single peer confirming an individual's judgment markedly decreased the influence of other—more distant—peers. Second, we develop a framework for modelling the cognitive processes underlying the integration of disparate social information, combining Bayesian updating with simple heuristics. Our model accurately accounts for observed adjustment strategies and reveals that people particularly heed social information that confirms personal judgments. Moreover, the model exposes strong inter-individual differences in strategy use. Third, using simulations, we explore the possible implications of identified strategies for belief updating more broadly. They show how confirmation effects can hamper the influence of disparate social information, exacerbate filter bubble effects and worsen group polarization. Overall, our results clarify what aspects of the social environment are, and are not, conducive to changing people's minds.

Introduction

The way in which individuals integrate personal and social information shapes a wide range of collective phenomena, including the spread of knowledge across social networks (1), the development of financial markets (2), political mobilization (3) and voting outcomes (4). By interacting with others and observing their behavior, individuals can often glean useful information helping them to, for example, rapidly adjust to new environments (5). However, with more and more of these interactions occurring online, and being controlled by algorithms that prioritize interactions between like-minded people (6), people can become cut off from information that might challenge their beliefs (7, 8). It has been argued that such dynamics may lead to stronger in-group consensus, and between-group polarization, in controversial matters ranging from same-sex marriage and vaccine safety to climate change (9–11). To counteract these dynamics, several recent projects (e.g., onesub.io and nuzzera.com) have focused on de-biasing news feeds and provide users with more balanced social information from disparate sources (for review see (12)). Yet, to effectively de-bias individuals we need a detailed understanding of the strategies individuals use when they are confronted with disparate social information coming from multiple sources.

Social information use is extensively studied across the biological and social sciences (13–28). In humans, social information use often involves changing one's mind after observing the behavior of other people (29–32). This process is commonly investigated using estimation tasks in which people are allowed to revise their first estimates after observing the estimate of a peer (e.g., (24, 33–36)). Studies using this approach give a detailed and quantified account of the effects of social cues on behavior, primarily focusing on how individuals incorporate a single piece of social information (29–31, 33, 36, 37). Studies considering multiple peers have mainly described the effect of the central tendency (e.g., the mean social information; (34, 38–40), but see (41)). In most real-world environments, however, people are confronted with multiple sources of social information at the same time, in various degrees of extremity. Currently, it is unclear how people integrate such disparate social information. Here we will address this issue in three steps.

First, we experimentally investigate how basic characteristics of the distribution of social information shape social information use. Specifically, we systematically manipulate the variance (reflecting the agreement among peers) and skewness (reflecting the clustering of peers close to or far away from the focal participant) of the distribution, while holding its mean constant. We show that the impact of social information strongly depends on its distribution. Disagreement among peers decreases their overall influence. Furthermore, the direction of the skew substantially alters the impact of social

information: participants adjust their first estimate considerably more when the majority of peers moderately agree with them and one peer strongly disagrees, compared to a situation in which a single peer strongly agrees with them, but the majority of peers strongly disagrees. This highlights the strong influence of confirmation effects.

Second, we introduce a formal model to explain the strategies underlying these adjustments. This model is informed by previous research on individuals' strategies for incorporating a single piece of social information. This research has identified three distinct strategies: (i) keeping one's initial beliefs, (ii) adopting the behavior of others, or (iii) 'compromising' between personal and social information (24, 30–32, 35, 42–48). We develop a modeling framework that extends these insights to situations with multiple social sources, accommodating both simple heuristics (keeping and adopting), and more complex strategies (compromising). Our model successfully recovers the main patterns in the observed data: it accurately predicts how the distribution of social information impacts the relative frequencies of adjustment strategies, and accounts for the strong between-individual heterogeneity in strategy use. Our modelling results reveal that social information receives more weight when it is in line with people's initial beliefs (reflecting confirmation effects), and when it is in close agreement with other social information (reflecting peer consensus).

Finally, we use our model to go beyond mere description, and use it to predict how identified adjustment strategies may shape belief dynamics in exemplary social environments that vary in the like-mindedness of peers. These simulations reveal how and when people's prioritizing of confirmatory social information can exacerbate filter bubble effects. Moreover, they illustrate how individual differences in confirmation effects can render beliefs to become more moderate (fostering group consensus) or more extreme (fueling group polarization).

Experimental Design

To examine how people integrate disparate information from multiple social sources, we used an adapted version of the BEAST (Berlin Estimate AdjuStment Task): a validated perceptual judgment task known to reliably measure individuals' social information use (Fig. 1 (37)). In the task, participants are repeatedly shown images of animal groups and have to estimate the number of animals (Fig. 1A,B). They then observe the estimates of three previous participants, and make a second estimate (Fig. 1C). The relative degree of adjustment quantifies an individual's social information use (Fig 1D (37)).

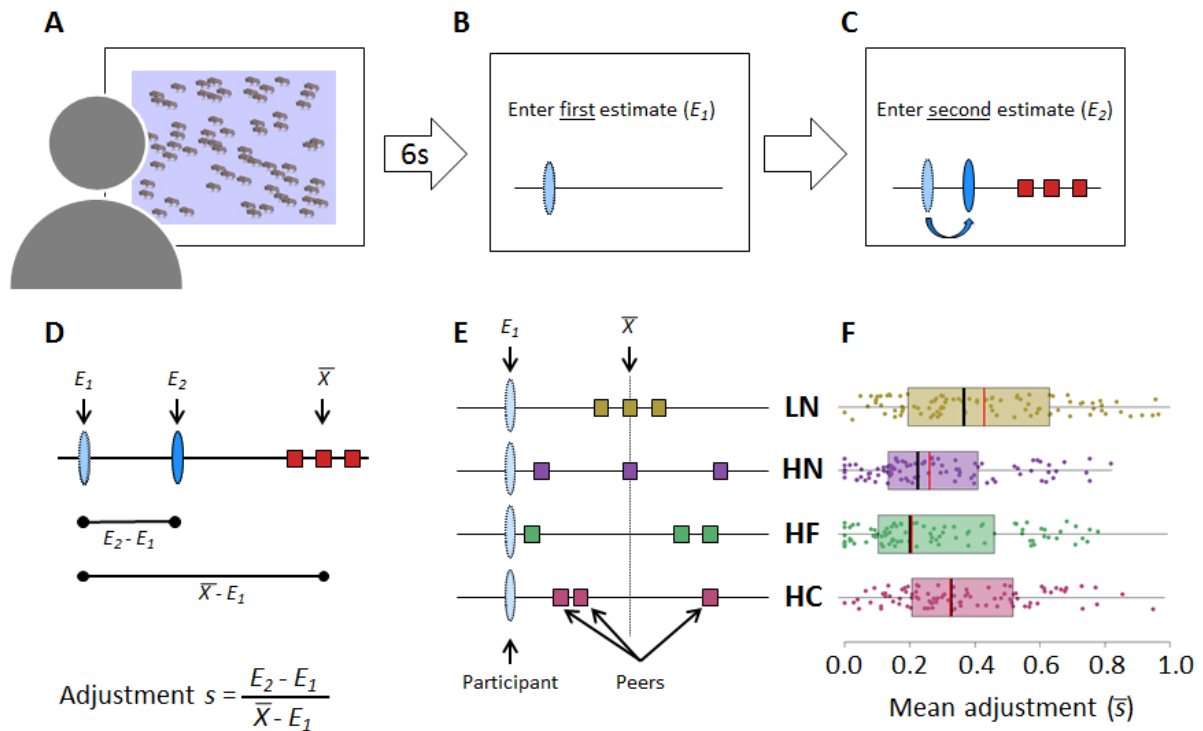


Fig. 1. Experimental paradigm and the impact of disparate social information. (A) Participants start with observing a group of animals for six seconds. (B) Next, they enter their first estimate of the total number of animals using a slider. (C) Then, they observe the estimates of three prerecorded peers (red squares), as well as their first estimate (light blue oval), and enter their second estimate (dark blue oval). (D) Social information use in a round (s) is calculated as the adjustment from the first estimate (E_1) to the second estimate (E_2), divided by the distance between the first estimate and the mean of the social information ($\bar{X}$). Rearranging the terms highlights that E_2 is an average of E_1 and $\bar{X}$, weighted by s : $E_2 = (1-s) \cdot E_1 + s \cdot \bar{X}$. (E) Across four conditions, we manipulated the distribution of social information (squares) relative to a participant's first estimate (oval). Across conditions, we manipulated the variance and skewness of the social information, while fixing the distance between the mean of the social information and the participant's first estimate (for details see Experimental Design). Peer estimates displayed either low variance (LN) or high variance but no skewness (HN), or high variance with a skewed distribution, with a cluster of two peers far from (HF), or close to (HC) E_1 . (F) Mean estimate shifts in each condition. Colored dots show participants' mean adjustments across the five rounds of each condition; $\bar{s}=1$ indicates a mean estimate shift to $\bar{X}$. Boxplots show the interquartile range (IQR), the median (black line), and the 1.5 IQR (whiskers). Red vertical lines show for each condition the predicted medians of the best-fitting model (see Cognitive Model). For strategies underlying mean estimate shifts across rounds, see Fig. 2.

We study participants' social information use across four conditions that systematically differ in variance and skewness, while controlling for the mean deviation from a participant's first estimate (Fig. 1E): (i) low variance, not skewed (LN); (ii) high variance, not skewed (HN); (iii) high variance, with a cluster of two peers relatively far from the participant's first estimate (HF); (iv) high variance, with a

cluster of two peers relatively close to a participant's first estimate (HC). These conditions encompass a broad range of distributions individuals may encounter when sampling their social environment. Across all conditions, the three pieces of social information always point in the same—and correct—direction (i.e., avoiding situations in which the social information brackets the personal estimate). Importantly, holding constant the relative mean deviation from a participant's first estimate across conditions implies that a participant weighting all peer estimates equally should make similar adjustments across all conditions.

Ninety-five participants completed thirty rounds of the judgment task. Each participant completed five rounds of each condition, presented in randomized order and interleaved with 10 'filler' rounds. Social information in the 'filler' rounds consisted of the estimates of three randomly drawn previous participants, ensuring that across all rounds, social information was sometimes higher and sometimes lower than a participant's first estimate, and sometimes bracketed the participant's personal estimate (see Methods for details). As a control, participants completed five additional rounds in which they did not observe the stimulus themselves, but only the estimates of four peers. The distribution of these peer estimates emulated the distributions of the four experimental conditions (i.e., one of each condition), plus a filler round. This enables us to compare how participants integrate personal and social information with this control condition, which reveals how participants aggregate a distribution of estimates in the absence of an own prior estimate (49, 50). Participants were recruited and performed the task online (see Methods).

Results

Experimental results. Participants' use of social information strongly depended on its distribution (Fig. 1F). Participants adjusted their estimates most when social information had low variance and no skewness (LN condition; Fig. 1F, yellow), shifting, on average, 41.5% towards the mean social information. In the high variance and no skew condition (HN condition; Fig. 1F, purple), average adjustments were credibly lower (mean adjustment: 29.0%; see Table S1 for statistics). Although adjustments in both conditions with skewed distributions were credibly lower than in the LN condition, the direction of the skew affected the relative adjustment: participants adjusted credibly more when two peers clustered relatively close by (HC condition; mean: 36.5%; Fig. 1F, red) rather than far from the participant (HF condition; mean: 27.9%; Fig. 1F, green). Overall, these results demonstrate that variance and skewness in peer behavior markedly affect social information use.

These results also show that average adjustments tended to be much smaller than what would be expected when participants weigh each of the three peer estimates as much as their own personal estimate (in which case they would adjust to the arithmetic mean of the four estimates, shifting 75% towards the mean social information in each condition). Moreover, participants substantially varied in their average adjustments, and these average adjustments strongly correlated across conditions (all pairwise Pearson correlations $r \geq 0.76$, $P < 0.001$; Fig. S1), indicating consistent inter-individual differences in social information use (see also below).

Figure 2 zooms in on the strategies underlying behavioral adjustments across rounds, differentiating between three distinct strategies: (1) *keeping* the first estimate, (2) *adopting* the estimate of one of the three peers, or (3) *compromising* between the first estimate and the peer estimates. The relative frequency of these strategies differed markedly between the four conditions (Fig. 2A-D; see Table S2 for statistics). When participants observed a single peer that closely agreed with them (HN and HF conditions; Fig. 2B,C), participants were more likely to either keep their first estimate, or to adopt the estimate of this near peer. When none of the peers was in close agreement with them (LN and HC conditions; Fig. 2A,D), participants were more likely to compromise, adjusting their estimate towards - but rarely beyond - the nearest peer. These results demonstrate that variance and skewness in peer behavior have strong effects on the strategies people use to integrate disparate social information. Figure 2E shows the frequency of strategies per participant across all conditions, illustrating that participants ranged from almost exclusively compromising, to exclusively keeping (see also Fig. S1), with compromising being the most frequent strategy.

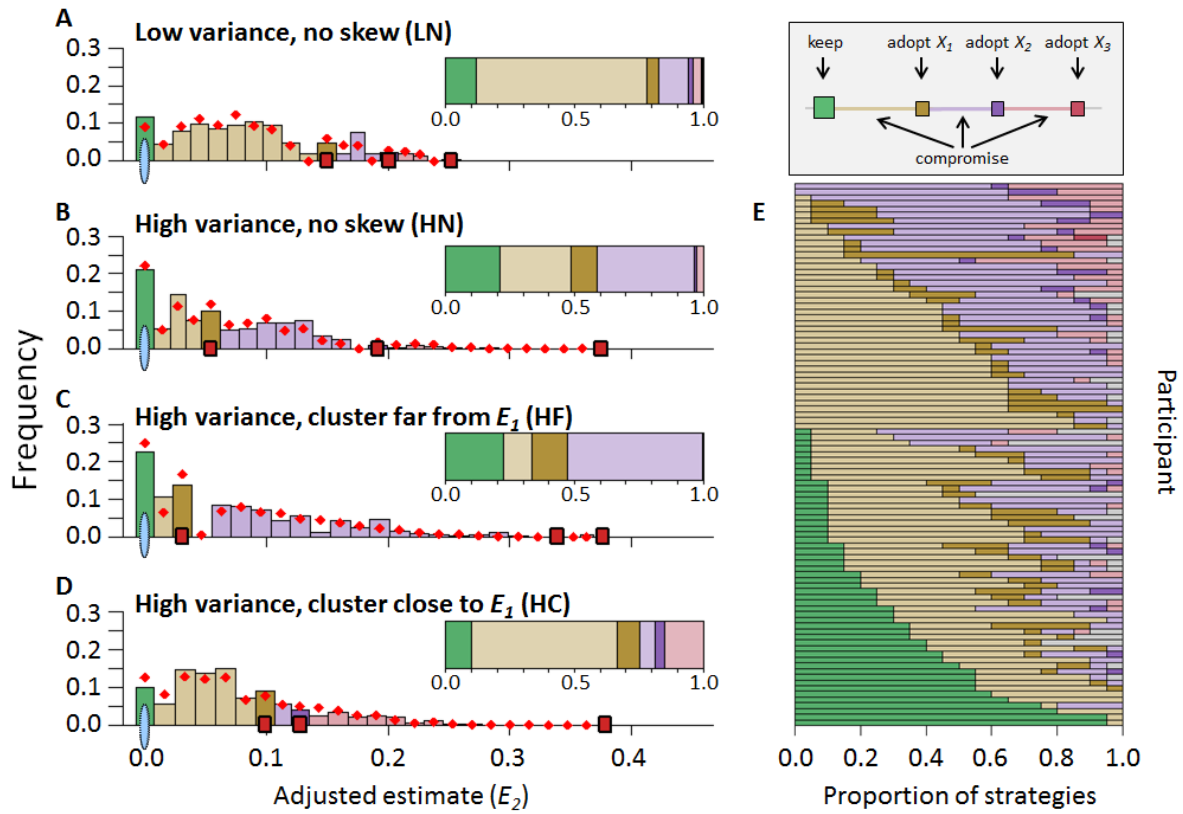


Fig 2. Adjustment strategies across conditions and participants (A-D) Bars indicate the observed distribution of adjustments in individual rounds, expressed as the fraction of a participant's first estimate (i.e., $|E_2 - E_1| / E_1$), per condition. The relative positions of the peer estimates (red squares) slightly varied across rounds (shown are their mean positions; see Methods for details). Bar and inset colors indicate the three strategies (i.e., keep, adopt, and compromise). Insets show the proportion of strategies per condition. Red diamonds show the predictions from the best cognitive model (see Cognitive Model), closely tracking the observed distributions. (E) The proportion of adjustment strategies per participant across all conditions. Participants (i.e., rows) are sorted according to their frequency of using the respective strategies.

In all four control conditions — in which participants did not observe the stimulus, but four peer estimates, emulating the four distributions of the experimental conditions — responses were close to the arithmetic mean of the four peer estimates (Figs. S2 & S3). Participants did, however, assign more weight to estimates closer to each other (Fig. S4). This indicates that the observed deviations from the arithmetic mean in the experimental conditions (Fig. 1F & Fig. 2A-D) are not due to an inability to integrate multiple pieces of information. Rather, the stark differences between the experimental and control conditions show that people down-weight social information that is more distant from their own first estimate, an effect known as 'egocentric discounting' (31, 32, 34–36, 41, 44–46, 51).

Cognitive Model. To investigate potential cognitive mechanisms underlying individuals' integration of disparate social information, we developed a set of models unifying heuristic (keeping and adopting) and more complex strategies (compromising; Fig. 3A). Based on our behavioral findings and previous literature, we assume that an individual's relative probabilities of using any adjustment strategy (keep, adopt or compromise) depends on the distance between their own first estimate and the estimate of the nearest peer (41, 51). We further assume that, when compromising, the weight of each of the peer estimates depends on its distance to an individual's own first estimate (32, 34–36), and its distance to other peer estimates (capturing a possible 'proximity effect'; Fig. S4; (41, 51)). These assumptions are reflected in four model features, defining the selection of (i) the keep heuristic, or (ii) the adopt heuristic, and, when compromising, the weighting of social information based on (iii) confirmation or (iv) proximity. We formalize these features below.

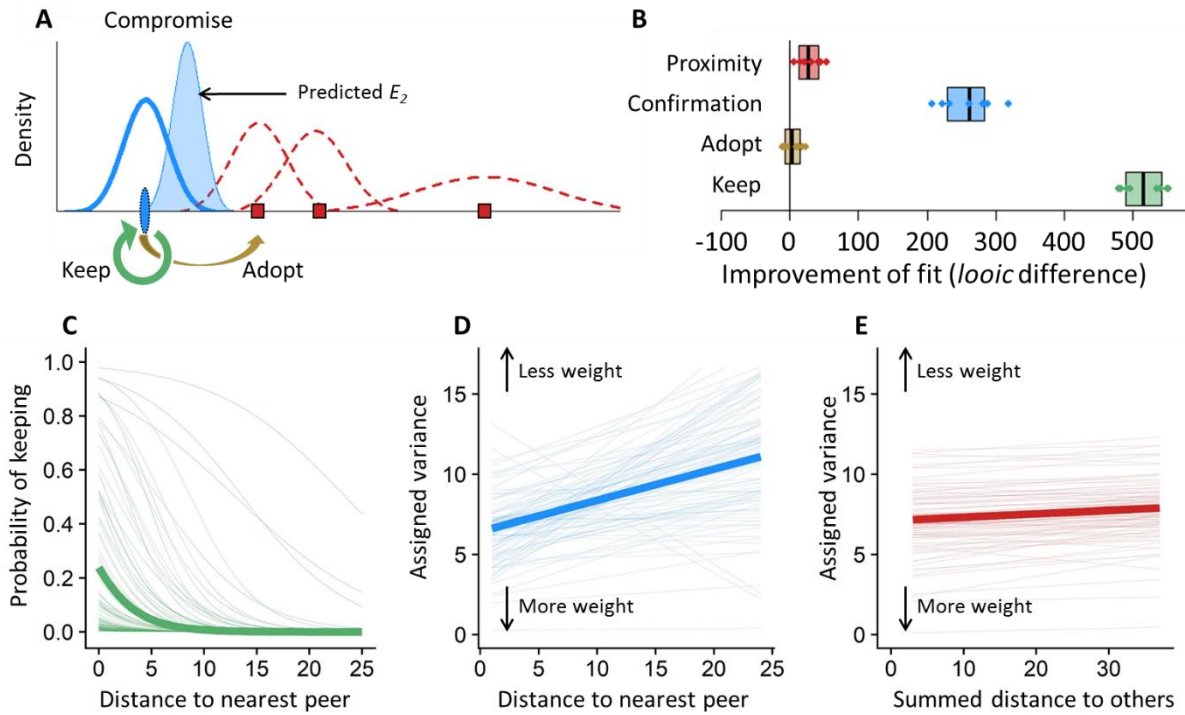


Fig. 3. Cognitive model of the integration of disparate social information. (A) We model participants' estimate adjustments by combining heuristics of keeping and adopting with compromising strategies. We model the probability of keeping the first estimate (green arrow) as a function of the distance between a participant's first estimate (E_1 ; blue oval) and the nearest peer (X_1). Similarly, the probability of adopting the nearest peer estimate (brown arrow) is a function of its distance to E_1 . Compromising entails taking a weighted average of E_1 and the peer estimates (X_i ; red squares): we predict a participant's second estimate (E_2 ; transparent blue) based on Bayesian updating using weighted means of E_1 and each X_i . Personal and social information are represented as probability density distributions with means at the observed estimates, and variances inversely related to their weight. Relative weights of social information depend on their distance to E_1 (i.e., degree of agreement with the participant; 'confirmation') and mean distance to other social information (i.e., degree of agreement with others; 'proximity'). (B) Improvement of fit for each of the model features based on $looic$ differences (Methods). Dots show fit improvements for pairs of models excluding and including each feature, and the box plots show the median improvement and IQR. (C-E) The effect of each feature included in the best-fitting model. (C) Fitted probability of keeping one's first estimate as a function of its distance to the nearest peer. (D and E) Variance assigned to peer estimates as a function of their distance to E_1 (D), and as their mean distance to other peers (E). Thin lines represent estimates for individuals and thick lines show group-level means.

We cast these assumptions into a Bayesian mixture model. First, individuals choose one of the three strategies with respective probabilities $P(\text{keep})$, $P(\text{adopt})$, and $P(\text{compromise})$ based on logistic functions (52). Formally, we define the probability to keep as $P(\text{keep}) = S(\alpha_{\text{keep}} + \beta_{\text{keep}} \cdot d_1)$, where α_{keep} and β_{keep} determine the intercept and slope of the standard logistic function (S), and d_1 is the

absolute distance between the first estimate and the nearest peer estimate: $d_1 = |E_1 - X_1|$. Likewise, we define the probability to adopt the estimate of the nearest peer as $P(\text{adopt}) = S(\alpha_{\text{adopt}} + \beta_{\text{adopt}} \cdot d_1)$. Note that we do not include the possibility to adopt the estimates of the peers that were not the nearest, as this rarely happened in our data (Fig. 2) and would make the model overly complicated.

The probability to compromise is given by: $1 - P(\text{keep}) - P(\text{adopt})$. Subsequently, compromising is modelled as a Bayesian updating process (40, 48, 53, 54): individuals weigh their first estimate (E_1) and the peer estimates (X_{1-3}) and generate an updated (posterior) belief (Fig. 3A). We assume that an individual's initial (prior) belief E_p about the number of animals (N) follows a discretized normal distribution centered around the first estimate E_1 . The uncertainty of the initial belief is captured in the variance of the distribution (σ_p^2): $p(E_p | N) \sim \text{Norm}(E_1, \sigma_p^2)$.

Likewise, we model social information (SI_s) as discretized normal distributions around peer estimates (X_s), with uncertainty σ_s^2 : $p(SI_s | N) \sim \text{Norm}(X_s, \sigma_s^2)$, where the subscript s indexes each peer estimate. Both σ_p^2 and σ_s^2 are free parameters indicating how much weight individuals assign to the estimates, with high values (i.e., more uncertainty) indicating less weight. When $\sigma_p^2 < \sigma_s^2$, participants assign more weight to their personal first estimate than to those of others.

Individuals further weigh each peer estimate based on the extent to which it confirms their own prior belief, and its proximity to the other two peer estimates. Confirmation-based weighting depends on the absolute distance between a peer estimate and an individual's first estimate ($d_s = |E_1 - X_s|$). Formally, this distance impacts the uncertainty assigned to the estimate: $\sigma_s^2 = \alpha_s + \beta_{\text{confirmation}} \cdot d_s$, where α_s and $\beta_{\text{confirmation}}$ respectively determine the intercept and slope of the weighting function. Positive values of $\beta_{\text{confirmation}}$ imply that peer estimates agreeing more with an individual's first estimate receive more weight.

Proximity-based weighting of a peer estimate depends on its summed absolute distance (e.g., $\tau_1 = |X_1 - X_2| + |X_1 - X_3|$) to the other two peer estimates: $\sigma_s^2 = \alpha_s + \beta_{\text{proximity}} \cdot \tau_s$. Positive values of $\beta_{\text{proximity}}$ imply that peer estimates agreeing more with other peer estimates receive more weight. When we consider both effects simultaneously, confirmation- and proximity-based weighting shape uncertainty σ_s^2 in an additive fashion. We use Bayes' rule to calculate updated beliefs about the number of animals by integrating E_p and SI_s (see Supplementary Methods). Note that compromising

results in a probability distribution for adjustments, which might result in instances of keeping or adopting.

We fit a series of models to simultaneously estimate the parameter values defining relative probabilities of using any adjustment strategies (keeping, adopting, or compromising), and confirmation- and proximity-based weighting. To account for individual differences in strategy use (Fig. 2E), we implement hierarchical Bayesian models (see Methods for implementation details). We evaluate the importance of the four model features ‘keep’, ‘adopt’, ‘confirmation’, and ‘proximity’ by calculating the leave-one-out cross-validation (looic (55)) of the 16 models comprising all possible combinations of these features (Table S3). Parameter recovery confirmed that our model fitting procedure yields robust and interpretable parameter values (Fig. S8; Supplementary Methods).

Figure 3B compares the fits of models including versus excluding each feature. It reveals that all but one considered feature reliably improve the model fit, with only ‘adopt’ marginally reducing predictive accuracy. Correspondingly, the best-fitting model includes the keep heuristic, and compromising based on both ‘confirmation’ and ‘proximity’ (Table S3). The parameter estimates from the best-fitting model are presented in Table S4, and the effects of its three features are illustrated in Fig. 3C-E. Overall, participants were more likely to apply a heuristic of ‘keeping’ when the nearest peer was in close agreement with them. Many participants frequently kept their first estimate when they observed a peer in close agreement with them, and this likelihood sharply dropped with increasing distance to the nearest peer ($\beta_{keep} < 0$; Fig. 3C).

When compromising, participants tended to weigh personal information more than social information ($\sigma_p^2 < \sigma_s^2$; Table S4). Importantly, participants assigned more weight to peers who more strongly agreed with them ($\beta_{confirmation} > 0$; Fig. 3D). This result is indicative of a confirmation effect (i.e., favoring information that affirms one’s beliefs). Furthermore, participants assigned more weight to peers who showed more agreement with other peers ($\beta_{proximity} > 0$; Fig. 3E). This proximity effect was, however, much smaller than the effect of ‘confirmation’ ($\beta_{proximity} < \beta_{confirmation}$; compare slopes in Fig. 3D vs E). For each of the features, the model detects substantial individual differences (indicated by the thin lines in Fig. 3C-E), thus capturing the high inter-individual differences in mean adjustment and strategy use (Figs. 2F, S1 and S5). Finally, we note that the absence of an effect of adopting the nearest peer estimate (Fig. 3B) may be because instances of adopting can be accounted for by adjustment through compromising.

Importantly, the best-fitting model closely predicts the mean adjustment across conditions (Fig. 1F; red vertical lines) as well as the distributions of adjustments in rounds across conditions (Fig. 2A-D; red diamonds). This shows that the model can account for the main patterns observed in our experimental conditions. Our model also accurately predicts participants' mean adjustment and keep probability in the 'filler' rounds (where peer estimates were randomly selected from the prerecorded pool and frequently bracketed the participant's first estimate; Fig. S5). Furthermore, the model can recover a commonly-observed phenomenon in estimation tasks, namely that mean adjustments are highest when social information is at intermediate distance from first estimates (Fig. S6 (34, 35)). Taken together, these results suggest that our model can generalize to cases that are qualitatively different from our experimental conditions (on which the model was trained).

Simulations. The identified strategies of social information use allow us to predict how they may shape belief shifts in settings where individuals encounter peers with various levels of like-mindedness. In the following, we use simulations to explore how the social environment and social information use may foster consensus, or, alternatively, lead to polarization. We simulate agents who, as in the experiment, observe other estimates and adjust their first estimate. Agents observe ten pieces of social information, across five qualitative different exemplary settings. We start with three settings in which an agent's first estimate is confirmed by either a *(i)* large majority, *(ii)* half of the peers, or *(iii)* only a small minority. We further simulate settings in which the focal agent is leaning towards one of two strongly disagreeing groups, and compare adjustment of agents with *(iv)* strong or *(v)* weak confirmation effect. In each setting, we simulate 1,000 agents whose adjustment strategies (i.e., their parameter setting) were sampled from the group-level distributions parameterized by the mean posterior estimates of the best-fitting model (see Methods for details).

Figure 4 shows the predicted adjustments for the five scenarios. *(i)* When agents predominantly observe social information that agrees with their prior beliefs (i.e., mirroring a 'filter bubble' (7, 56)), they predominantly keep their first estimate or, at most, make a very small adjustment (Fig. 4A). *(ii)* Even when only half of the peers agree with them—and the other half strongly disagrees (reflecting a scenario typical for a de-biasing attempt (12))—agents only shift little, remaining far away from the global mean estimate (Fig. 4B). This suggests that even regular exposure to opposing information (e.g., from outside one's filter bubble) is unlikely to lead to substantial adjustments. *(iii)* Even when only a small minority agrees with them, agents are still prone to keep their first estimate, although adjustments towards the majority become more substantial (Fig. 4C). Overall, these simulation results illustrate how confirmation effects can curb belief updating: confirmatory social information

reinforces people's prior beliefs and prompts them to retain these beliefs, even if they reflect minority views.

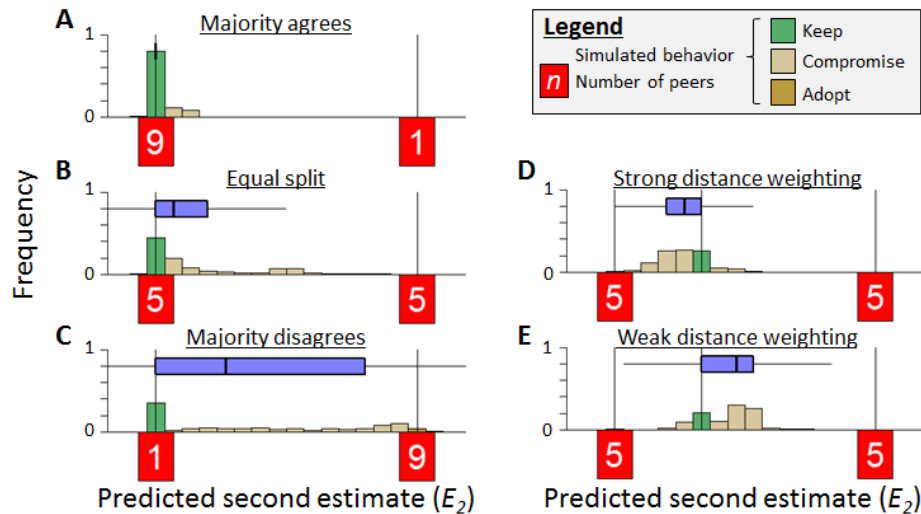


Fig. 4. Simulated adjustments of agents observing ten pieces of social information in varying compositions. In the simulations, peer estimates were always Low or High (vertical black lines; see Methods for details). Numbers in the red boxes correspond to the number of peers in each category. In (A-C) The agent's first estimate is also Low. Agents likely keep their first estimate when they observe (A) a large majority agreeing with them, (B) half of the peers agreeing with them, and (C) only a small minority agreeing with them, though in the latter case agents shifted substantially more. In (D-E), peers are equally split, and the agent's first estimate is in between the two clusters of peers, but closer to one cluster than the other. (D) Agents with a strong confirmation effect (sampled from the upper 50% of the distribution; cf. steepest slopes in Fig. 3D) tend to adjust towards the local extreme estimate. (E) Agents showing a weak confirmation effect (lower 50% of the distribution) tend to adjust towards the global mean estimate. In all panels, vertical bars show distributions of predicted adjustments across 1,000 simulations. The horizontal boxplots summarize these distributions, showing the median, IQR and the 1.5 IQR (whiskers).

To further examine the possible implications of individual differences in confirmation effects, we simulated adjustments for agents with high and low levels of this effect (i.e., individuals with steep and shallow slopes in Fig. 3D). Agents were initially located in between two clusters of peers, but slightly closer to one of the clusters. In this setting, agents with strong confirmation effects tend to adjust towards the local cluster—moving away from the global mean (Fig. 4D). Weak confirmation effects tend to lead to adjustments towards the global mean estimate (Fig. 4E). These results suggest that strong confirmation effects can drive people to more extreme views, and may increase polarization over time.

Discussion

Social information is increasingly consumed online, and often being controlled by algorithms that prioritize like-minded sources over others (7, 8). It has been argued that this prioritization contributes to increased polarization (9–11). A prominent approach to counteract polarization and promote consensus is to provide users with more balanced social information from disparate sources (12). Although interventions like this have a strong intuitive appeal, it is hitherto largely unknown how people integrate information from multiple social sources that (dis)agree with oneself and each other.

This paper makes three novel contributions. First, we experimentally show that the impact of multiple sources of social information strongly depends on its distribution. Increased variance in social information reduces participants' adjustments, and skewness decreases adjustments if a single peer confirms their first estimate. Second, our cognitive model provides a unified account for how people integrate disparate social information, showing that people rely on a combination of simple heuristics of keeping their initial beliefs and compromising towards social information. The model captures how the weight of social information is independently determined by both its degree of confirmation of people's initial beliefs, and its proximity to other pieces of social information. Third and finally, the model made accurate out-of-sample predictions of adjustments in qualitatively different judgment situations, and our simulations illustrate how prioritizing confirmatory social information may exacerbate filter bubble effects and even lead individuals to take up more extreme beliefs.

Overall, people assigned more weight to their personal initial beliefs than to social information (Fig. 1F, Fig. 2A-D; Fig. S3; Fig. S4; Table S4). It may seem somewhat puzzling why they would do so in a task in which social information consists of judgments of people incentivised to accurately solve the same problem. Indeed, there is no reason to assume that one's first estimate would be more accurate than those of others. One reason to prioritize personal estimates is the lack of access to others' reasons for holding their beliefs, which may lead participants to discount social information, especially when it is very distinct from one's personal beliefs (36, 57) (but see (49)). Interestingly, while consistent social information is weighted more when people compromise ('confirmation-based weighting'; Fig. 3D), it also increases the chance that people simply keep their initial beliefs (Fig. 3C). The combined result of these effects is that social information tends to have the strongest impact on overall adjustments when it is at intermediate distance (Fig. S6). Our results reveal that people also weight social information based on its consistency with other social information ('proximity-based weighting'; Fig. 3E; Fig. S4). One rationale for prioritizing social information consistent with other social information is

relatively straightforward: when people are motivated and able to make valid judgments, agreement among peers reliably signals accuracy (22, 58). The concerted action of the three mechanisms (heuristics of keeping, and compromising based on confirmation and peer proximity) explain our experimental observations that the impact of social information strongly depends on the variance and skewness of its distribution.

Our cognitive model provides a unified framework that elucidates how people combine a heuristic strategy (i.e., keeping when close) and compromising (based on weighted averaging) to integrate personal and social information. Model comparison revealed that assuming a separate heuristic for ‘adopting’ is not necessary, simply because compromising can predict that an updated estimate coincides with the nearest piece of social information. Moreover, our hierarchical modeling approach allowed for accounting for individual differences in social information use, a regularly observed but poorly understood phenomenon (59–61). In sum, our model provides a detailed account of people’s adjustment strategies, contributing to understanding the hitherto understudied computational and cognitive mechanisms underlying social information use (62).

Our model’s accurate predictions of behavior in the ‘filler’ rounds underscores its ability to go beyond mere redescription of the empirical data, and suggests that our model can be generalized to settings that are qualitatively different from our experimental conditions (Fig. S5; Fig. S6). Our simulations go beyond the limited set of distributions of social information studied in our experiments, generating predictions about how distributions resembling important real-world settings (e.g. filter bubbles) shape social information use. These simulations predict that disparate social information changes people’s minds only to a limited degree, even when this social information signals that people hold minority views (Fig. 4). More importantly, under certain conditions, observing balanced social information can even lead individuals to take more extreme views (Fig. 4D,E). These findings have direct implications for interventions. For instance, they suggest that efforts to de-bias online information that present people with balanced views (12) might not suffice to break filter bubble effects and dynamics of polarization.

The current study provides a robust template for understanding social information use in a range of (complex) social environments that people encounter in their day-to-day lives. Future empirical work should test the predictions of our simulations, as well as the extent to which our findings generalise to other domains of decision making. Our results demonstrate that confirmation effects can strongly reduce people’s willingness to change their minds even in the minimal setting of a task with an

objectively correct solution. It seems plausible that confirmation effects - and their polarizing consequences - are even stronger in many important real-world contexts involving emotive, moral or political issues. In these (often controversial) contexts, the integration of disparate social information may be further hampered due to 'motivated reasoning' (63) or when observed peers belong to an out-group (64, 65). Finally, in contrast to our experiment, people are often confronted with different sources of information sequentially. The current experimental design and modeling framework are flexible enough to include adaptations to accommodate each of these elements.

To conclude, our findings contribute to a growing literature on how people integrate social information to update their beliefs. We go beyond previous experimental work on the effects of social information (which focused on single social cues or the mean of multiple cues) by showing that the variance and skewness of its distribution strongly modulates its impact on behavior. Our cognitive model provides a detailed picture of the cognitive mechanisms that underlie the integration of disparate social information, highlighting the role of heuristics of keeping one's initial beliefs, and the importance of proximity- and confirmation-based weighting. Finally, our simulations illustrate how confirmation effects can markedly exacerbate filter bubble dynamics and polarization. We anticipate that these findings will provide a useful point of departure for future work aiming to understand the nature of human social information use and its implications for group dynamics, and to inform interventions to effectively de-bias individuals and help them forming accurate beliefs about the world.

Methods

Behavioral experiment. We recruited 100 participants from Amazon Mechanical Turk (MTurk), restricted to US citizens, and to MTurk workers with an approval rate of at least 95%. By clicking the link to the experimental pages, participants confirmed informed consent. Five participants dropped out during the task and did not receive any payment, resulting in 95 participants (57% male, mean age: 35.8; s.d.=10.7). The online experiment was programmed in LIONESS Lab (66). All code associated with this paper (i.e., experimental software, cognitive model, simulations, and analyses) is publicly available in the public OSF repository <https://osf.io/rmcuy/>. Ethical approval was obtained from the Institutional Review Board of the Max Planck Institute for Human Development Berlin (ARC 2017/18).

The experimental task is based on a validated perceptual judgment paradigm for quantifying social information use (BEAST (37); Fig. 1A-C). In each of 30 rounds (5 per experimental condition, plus 10 'filler' rounds with randomly selected social information, see Supplementary Methods; section

‘Definition of experimental conditions’), participants observed an image with 50-100 animals for 6 seconds and had to estimate how many animals there were. They entered their estimates with a slider limited from 1 to 150 (for screenshots, see Supplementary Information, section Experimental Materials). After submitting their first estimate (E_1), they observed the estimates of three other participants (X_1, X_2, X_3) who had completed the same task before but without receiving social information. After observing the social information, participants provided a second estimate (E_2). Participants were rewarded for accuracy, earning 100 points if their estimate was exactly correct (both for E_1 and E_2). For each animal they were off, five points were subtracted (but earnings in a round could not drop below zero). At the end of the session, one decision was randomly chosen from each of the experimental ‘blocks’ (see Supplementary Methods) for bonus payment (100 points = \$1.00), which came on top of a flat fee of \$4.50. Total earnings ranged from \$4.50 to \$7.00 (average \$5.50). Participants took, on average, 35 minutes, resulting in an hourly wage of \$9.50. Experimental sessions ended with a short questionnaire in which we recorded participants’ age and gender, and measured individualism (Triandis & Gelfand, 1998), social conformity (Mehrabian and Stefl 1995), and resistance to peer influence (Steinberg & Monahan 2007); see caption of Fig. S1.

To study the effects of disparate social information on behavior, we used four experimental conditions (Fig. 1E). Across conditions, we systematically manipulated the variance and skewness of the distributions of social information, while keeping the distance between the mean social information ($\bar{X}$) and E_1 constant. To achieve experimental control without deception, we first recorded a large ($N=100$) pool of estimates made by MTurkers for each image shown in the main experiment. In a given trial, the three pieces of social information were selected based on the participant's first estimate and the experimental condition assigned to that trial (for implementation details, see Supplementary Methods). This procedure resulted in clearly defined experimental conditions (Fig. S7). We randomly shuffled the order of experimental conditions across rounds and held this order fixed for all participants.

Cognitive Model. We analysed the cognitive model with a hierarchical Bayesian MCMC technique implemented with *RStan* in *R* (67, 68); for implementation of the sampler and the hierarchical model structure, see Supplementary Methods). To investigate the predictive power of the four model features ‘keep’, ‘adopt’, ‘confirmation’, and ‘proximity’ we calculated the *looic* (55) of the 16 model variants comprising all possible combinations of these features (Table S3). We quantified the importance of a feature by calculating the average reduction of the *looic* when the feature was included rather than excluded (Fig. 2B). In the main text (e.g. Fig. 3C-E) we focus on the results of the best-fitting model, i.e., the model with the lowest *looic* (see Table S3; Supplementary Methods).

We generated the predictions of the best-fitting model (red lines and diamonds in Fig. 1F and Fig. 2, respectively) by calculating the probability density function for each participant in each round. This density function was based on the mean posterior parameter estimates for the participant (see Table S4 for characteristics of the parameter group-level distributions), their experimentally observed first estimate (E_1) and the displayed social information (X_s) in that round. To account for stochasticity, the model predictions in Figure 1F (red line) and Figure 2 (red diamonds) are based on 10 samples of estimates from each density function. To analyse the prediction of the best-fitting model for the experimental conditions and ‘filler’ rounds, we once sampled from each density function of each round and participant and calculated the actual and predicted individual-level mean adjustment and keep proportions (Fig. S5). To verify that the model parameters are distinguishable and identifiable, we conducted extensive parameter recovery analyses (Fig. S8). The results (reported in detail in the Supplementary Methods) confirm that the model allows for reliably recovering all of its parameters.

Simulations. In each of the settings shown in Fig. 4, agents were endowed with adjustment strategies whose parameter values were sampled from the group-level distributions from the best-fitting model (Table S4), assuming no correlations between the parameters. Inspection of the joint posterior distributions of means shows that this assumption is generally justified (for details, see Table S5; Fig. S9; sampling from the individual-level parameters from our experiment does not lead to observable differences in simulation outcomes shown in Fig. 4). For each agent, we assumed their first estimate to be 50, and simulated their adjustment given the social information in the respective social environment. For Fig. 4, the social environment consisted of ten pieces of social information, either agreeing (i.e., an estimate of 50) or disagreeing (i.e., an estimate of 65) with the focal agent. For Fig. 4D and 4E we sampled the parameters for confirmation-based weighting from the higher and lower half of the group-level distribution of $\beta_{confirmation}$, respectively. The individuals’ first estimate in these settings was 55 (that is, in between the two clusters of estimates, but closer to the lower cluster).

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Footnotes

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References

1. E. Bakshy, I. Rosenn, C. Marlow, L. Adamic, The Role of Social Networks in Information Diffusion in *Proceedings of the 21st International Conference on World Wide Web, WWW '12.*, (ACM, 2012), pp. 519–528.
2. A. Devenow, I. Welch, Rational herding in financial economics. *European Economic Review* **40**, 603–615 (1996).
3. R. M. Bond, *et al.*, A 61-million-person experiment in social influence and political mobilization. *Nature* **489**, 295–298 (2012).
4. A. J. Stewart, *et al.*, Information gerrymandering and undemocratic decisions. *Nature* **573**, 117–121 (2019).
5. R. Boyd, P. J. Richerson, J. Henrich, The cultural niche: Why social learning is essential for human adaptation. *Proceedings of the National Academy of Sciences* **108**, 10918–10925 (2011).
6. L. Rokach, F. Ricci, B. Shapira, *Recommender systems handbook* (Springer, 2015).
7. E. Pariser, *The filter bubble: What the Internet is hiding from you* (Penguin UK, 2011).
8. C. Sunstein, R. 2007. *Republic. com 2.0* (Princeton, NJ: Princeton University Press, 2003).
9. T. Deryugina, O. Shurchkov, The effect of information provision on public consensus about climate change. *PloS one* **11**, e0151469 (2016).
10. J. R. Kerr, M. S. Wilson, Perceptions of scientific consensus do not predict later beliefs about the reality of climate change: A test of the gateway belief model using cross-lagged panel analysis. *Journal of Environmental Psychology* **59**, 107–110 (2018).
11. S. Lewandowsky, G. E. Gignac, S. Vaughan, The pivotal role of perceived scientific consensus in acceptance of science. *Nature Climate Change* **3**, 399–404 (2013).
12. E. Bozdag, J. van den Hoven, Breaking the filter bubble: democracy and design. *Ethics and Information Technology* **17**, 249–265 (2015).
13. P. P. Analytis, D. Barkoczi, S. M. Herzog, Social learning strategies for matters of taste. *Nature Human Behaviour* **2**, 415 (2018).
14. P. van den Berg, L. Molleman, F. J. Weissing, Focus on the success of others leads to selfish behavior. *PNAS* **112**, 2912–2917 (2015).
15. R. Bond, P. B. Smith, Culture and conformity: A meta-analysis of studies using Asch's (1952b, 1956) line judgment task. *Psychological bulletin* **119**, 111 (1996).
16. R. Boyd, P. J. Richerson, *Culture and the evolutionary process* (University of Chicago Press, 1985).
17. E. Danchin, *et al.*, Cultural flies: Conformist social learning in fruitflies predicts long-lasting mate-choice traditions. *Science* **362**, 1025–1030 (2018).
18. M. Derex, J.-F. Bonnefon, R. Boyd, A. Mesoudi, Causal understanding is not necessary for the improvement of culturally evolving technology. *Nature Human Behaviour* **3**, 446 (2019).
19. W. Hoppitt, K. N. Laland, *Social Learning: An Introduction to Mechanisms, Methods, and Models* (Princeton University Press, 2013).
20. R. L. Kendal, *et al.*, Social learning strategies: Bridge-building between fields. *Trends in Cognitive Sciences* **22**, 651–665 (2018).
21. R. H. J. M. Kurvers, *et al.*, Personality predicts the use of social information. *Ecology Letters* **13**, 829–837 (2010).
22. H. Mercier, O. Morin, Majority rules: how good are we at aggregating convergent opinions? *Evolutionary Human Sciences* **1** (2019).
23. A. Mesoudi, *Cultural evolution: how Darwinian theory can explain human culture and synthesize the social sciences* (University of Chicago Press, 2011).

24. M. Moussaïd, S. M. Herzog, J. E. Kämmer, R. Hertwig, Reach and speed of judgment propagation in the laboratory. *Proceedings of the National Academy of Sciences* **114**, 201611998 (2017).
25. S. E. Page, *The Difference: How the Power of Diversity Creates Better Groups, Firms, Schools, and Societies - New Edition* (Princeton University Press, 2008).
26. L. Rendell, *et al.*, Why copy others? Insights from the social learning strategies tournament. *Science* **328**, 208–213 (2010).
27. J. Surowiecki, *The wisdom of crowds: Why the many are smarter than the few and how collective wisdom shapes business, economies, societies, and nations* (Doubleday & Co, 2004).
28. A. N. Tump, M. Wolf, J. Krause, R. H. J. M. Kurvers, Individuals fail to reap the collective benefits of diversity because of over-reliance on personal information. *Journal of The Royal Society Interface* **15**, 20180155 (2018).
29. P. Bednarik, T. Schultze, The effectiveness of imperfect weighting in advice taking. *Judgment and Decision Making* **10**, 265–276 (2015).
30. J. B. Soll, R. P. Larrick, Strategies for revising judgment: How (and how well) people use others' opinions. *Journal of Experimental Psychology: Learning, Memory, and Cognition* **35**, 780–805 (2009).
31. I. Yaniv, Weighting and Trimming: Heuristics for Aggregating Judgments under Uncertainty. *Organizational Behavior and Human Decision Processes* **69**, 237–249 (1997).
32. I. Yaniv, Receiving other people's advice: Influence and benefit. *Organizational Behavior and Human Decision Processes* **93**, 1–13 (2004).
33. S. Bonaccio, R. S. Dalal, Advice taking and decision-making: An integrative literature review, and implications for the organizational sciences. *Organizational Behavior and Human Decision Processes* **101**, 127–151 (2006).
34. B. Jayles, *et al.*, How social information can improve estimation accuracy in human groups. *Proceedings of the National Academy of Sciences*, 201703695 (2017).
35. M. Moussaïd, J. E. Kämmer, P. P. Analytis, H. Neth, Social influence and the collective dynamics of opinion formation. *PloS one* **8**, e78433 (2013).
36. I. Yaniv, E. Kleinberger, Advice Taking in Decision Making: Egocentric Discounting and Reputation Formation. *Organizational Behavior and Human Decision Processes* **83**, 260–281 (2000).
37. L. Molleman, R. H. J. M. Kurvers, W. van den Bos, Unleashing the BEAST: a brief measure of human social information use. *Evolution and Human Behavior* **40**, 492-499 (2019) <https://doi.org/10.1016/j.evolhumbehav.2019.06.005> (July 23, 2019).
38. R. P. Larrick, J. B. Soll, Intuitions About Combining Opinions: Misappreciation of the Averaging Principle. *Management Science* **52**, 111–127 (2006).
39. A. E. Mannes, Are We Wise About the Wisdom of Crowds? The Use of Group Judgments in Belief Revision. *Management Science* **55**, 1267–1279 (2009).
40. S. A. Park, S. Goïame, D. A. O'Connor, J.-C. Dreher, Integration of individual and social information for decision-making in groups of different sizes. *PLOS Biology* **15**, e2001958 (2017).
41. I. Yaniv, M. Milyavsky, Using advice from multiple sources to revise and improve judgments. *Organizational Behavior and Human Decision Processes* **103**, 104–120 (2007).
42. L. Aitchison, D. Bang, B. Bahrami, P. E. Latham, Doubly Bayesian Analysis of Confidence in Perceptual Decision-Making. *PLOS Computational Biology* **11**, e1004519 (2015).
43. B. Bahrami, *et al.*, Optimally Interacting Minds. *Science* **329**, 1081–1085 (2010).

44. D. V. Budescu, A. K. Rantilla, H.-T. Yu, T. M. Karelitz, The effects of asymmetry among advisors on the aggregation of their opinions. *Organizational Behavior and Human Decision Processes* **90**, 178–194 (2003).
45. D. V. Budescu, H.-T. Yu, Aggregation of opinions based on correlated cues and advisors. *Journal of Behavioral Decision Making* **20**, 153–177 (2007).
46. C. Harries, I. Yaniv, N. Harvey, Combining Advice: The Weight of a Dissenting Opinion in the Consensus. *Journal of Behavioral Decision Making* **17**, 333–348 (2004).
47. N. Shea, *et al.*, Supra-personal cognitive control and metacognition. *Trends in Cognitive Sciences* **18**, 186–193 (2014).
48. U. Toelch, R. J. Dolan, Informational and normative influences in conformity from a neurocomputational perspective. *Trends in cognitive sciences* **19**, 579–589 (2015).
49. J. B. Soll, A. E. Mannes, Judgmental aggregation strategies depend on whether the self is involved. *International Journal of Forecasting* **27**, 81–102 (2011).
50. I. Yaniv, S. Choshen-Hillel, Exploiting the wisdom of others to make better decisions: Suspending judgment reduces egocentrism and increases accuracy. *Journal of Behavioral Decision Making* **25**, 427–434 (2012).
51. T. Schultze, A.-F. Rakotoarisoa, S. Schulz-Hard, Effects of distance between initial estimates and advice on advice utilization. *Judgment and Decision making* **10**, 144–172 (2015).
52. B. Scheibehenne, J. Rieskamp, E.-J. Wagenmakers, Testing adaptive toolbox models: A Bayesian hierarchical approach. *Psychological review* **120**, 39 (2013).
53. D. Adjodah, Y. Leng, S. K. Chong, P. Krafft, A. Pentland, Social Bayesian Learning in the Wisdom of the Crowd. *arXiv:1712.09960 [physics]* (2017) (July 30, 2019).
54. D. Bang, C. Frith, Making better decisions in groups. *Royal Society Open Science* **4**, 170193 (2017).
55. A. Vehtari, A. Gelman, J. Gabry, loo: Efficient leave-one-out cross-validation and WAIC for Bayesian models. *R package version 1* (2016).
56. E. Bakshy, S. Messing, L. A. Adamic, Exposure to ideologically diverse news and opinion on Facebook. *Science* **348**, 1130–1132 (2015).
57. F. Ravazzolo, Ø. Røisland, Why do people place lower weight on advice far from their own initial opinion? *Economics Letters* **112**, 63–66 (2011).
58. R. H. Kurvers, *et al.*, How to detect high-performing individuals and groups: Decision similarity predicts accuracy. *Science Advances* **5**, eaaw9011 (2019).
59. C. Efferson, R. Lalive, P. J. Richerson, R. McElreath, M. Lubell, Conformists and mavericks: the empirics of frequency-dependent cultural transmission. *Evolution and Human Behavior* **29**, 56–64 (2008).
60. A. Mesoudi, L. Chang, S. R. Dall, A. Thornton, The evolution of individual and cultural variation in social learning. *Trends in Ecology & Evolution*, 215–225 (2016).
61. L. Molleman, P. van den Berg, F. J. Weissing, Consistent individual differences in human social learning strategies. *Nature Communications* **5**, 3570 (2014).
62. C. Heyes, Blackboxing: social learning strategies and cultural evolution. *Phil. Trans. R. Soc. B* **371**, 20150369 (2016).
63. C. A. Bail, *et al.*, Exposure to opposing views on social media can increase political polarization. *PNAS* **115**, 9216–9221 (2018).
64. D. Guilbeault, J. Becker, D. Centola, Social learning and partisan bias in the interpretation of climate trends. *Proceedings of the National Academy of Sciences* **115**, 9714–9719 (2018).

65. A. M. Votruba, V. S. Y. Kwan, Disagreeing on Whether Agreement Is Persuasive: Perceptions of Expert Group Decisions. *PLOS ONE* **10**, e0121426 (2015).
66. M. Giamattei, L. Molleman, K. Seyed Yahosseini, S. Gaechter, "LIONESS Lab – a free web-based platform for conducting interactive experiments online. Available at SSRN: <https://ssrn.com/abstract=3329384> or <http://dx.doi.org/10.2139/ssrn.3329384>" (Social Science Research Network, 2019) (February 16, 2019).
67. R Core Team, *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2015).
68. Stan Development Team, *RStan: the R Interface to Stan. R package version 2.17.3* (2018).
69. S. Nakagawa, H. Schielzeth, Repeatability for Gaussian and non-Gaussian data: a practical guide for biologists. *Biological Reviews* **85**, 935–956 (2010).
70. P.-C. Bürkner, brms : An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software* **80** (2017).
71. C. Dutang, P. Savicky, randtoolbox: Generating and testing random numbers. *R package*, 67 (2013).

Supplementary Information

For 'Strategies for integrating disparate social information', by Lucas Molleman, Alan N. Tump, Andrea Gradassi, Bertrand Jayles, Stefan M. Herzog, Ralf H.J.M. Kurvers and Wouter van den Bos

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Supplementary Figures

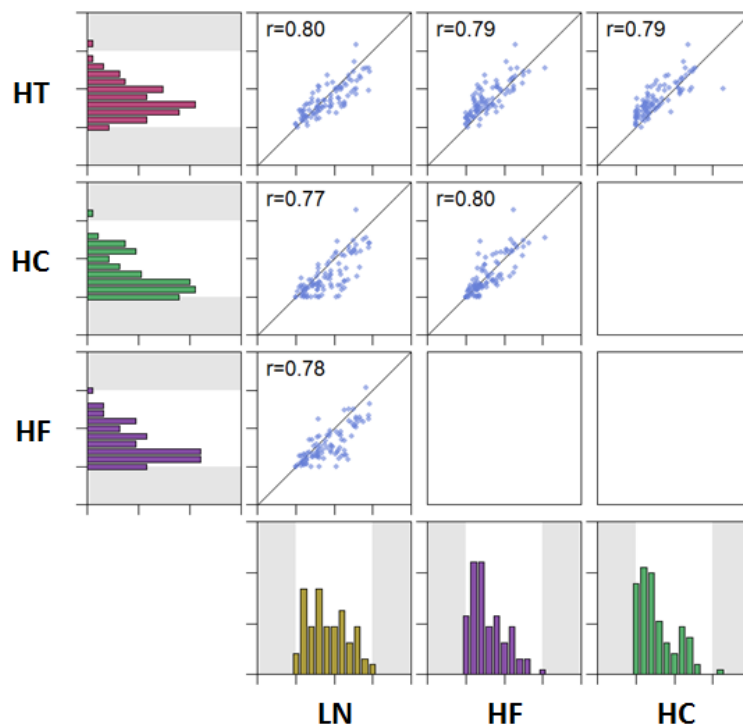


Fig. S1. Distributions of participants' mean adjustment and correlations across the four experimental conditions. Outer histogram panels show frequency distributions of participants' mean adjustments ($\bar{s}$) towards the mean social information for each condition (LN: low variance, no skewness; HN: high variance, no skewness; HF: high variance, cluster of two peers relatively far from the participants first estimate; HC: high variance, cluster of two peers relatively near to the participant's first estimate). Left (right) grey areas in each panel represent values of $\bar{s}$ below 0 (above 1). Across conditions, almost all participants had, on average, an $\bar{s}$ value between 0 and 1, implying they did not adjust away from the social information (i.e., $\bar{s} < 0$), nor adjust beyond the mean social information (i.e., $\bar{s} > 1$). Scatter plots show correlations between participants' mean $\bar{s}$ across the respective conditions. Dots represent participants, and the top left of each panel shows Pearson's r . Overall, we find strong correlations between participants' mean adjustments across conditions (all Pearson correlations $P < 0.001$), indicating consistent inter-individual differences in social information use. Furthermore, participants' mean adjustments averaged across all conditions correlated in the expected directions with self-reported questionnaire scales measuring conformity (Pearson's $r = 0.258$, $CI = [0.059, 0.436]$), individualism ($r = -0.204$, $CI = [-0.389, -0.002]$), and resistance to peer influence ($r = -0.233$, $CI = [-0.415, -0.033]$), in line with previous findings using this paradigm (37).

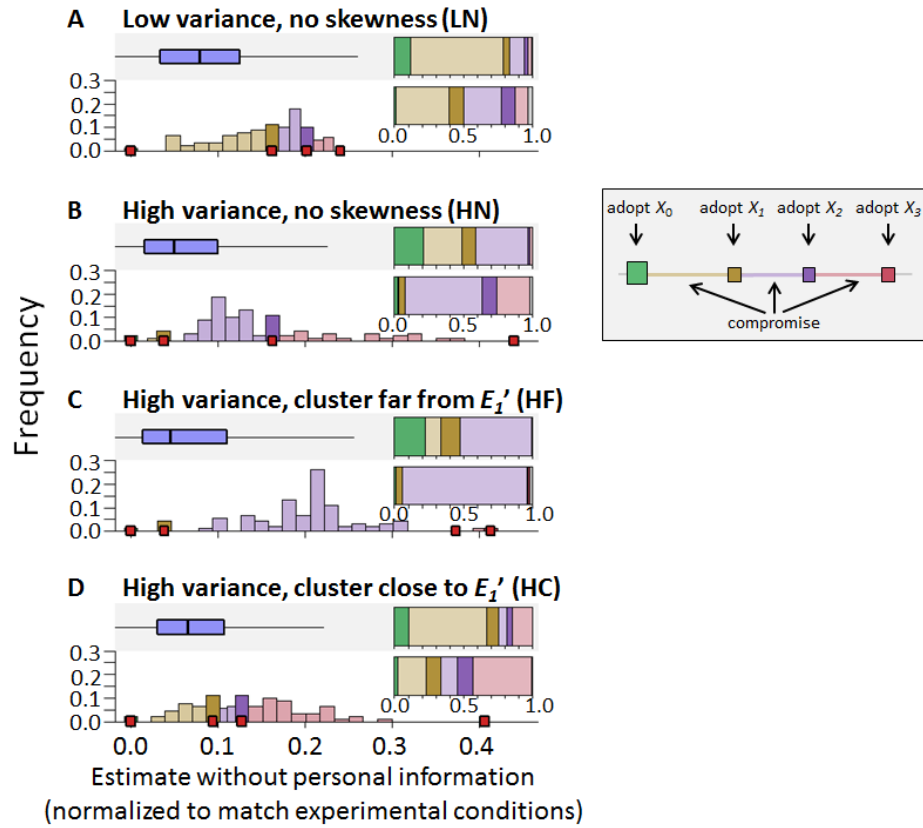


Fig. S2. Adjustments in control conditions in which participants did *not* observe the stimulus, but only observed four peer estimates. Trials were created by first drawing one estimate (E_1') at random from a pool of prerecorded estimates. Then, three additional prerecorded estimates were drawn analogous to the four experimental conditions (see Methods). Before plotting, data was normalized in the same way as Fig. 2A-D, taking E_1' as the reference point. Histograms show distributions of estimates in the same colour coding as Fig. 2A-D. For comparison, in the grey background on top of each panel we summarize behavior in conditions where participants *did* observe the stimulus (cf. Fig. 2A-D; blue boxplot: distributions of relative estimates; insets: relative frequencies of each of the qualitative cases).

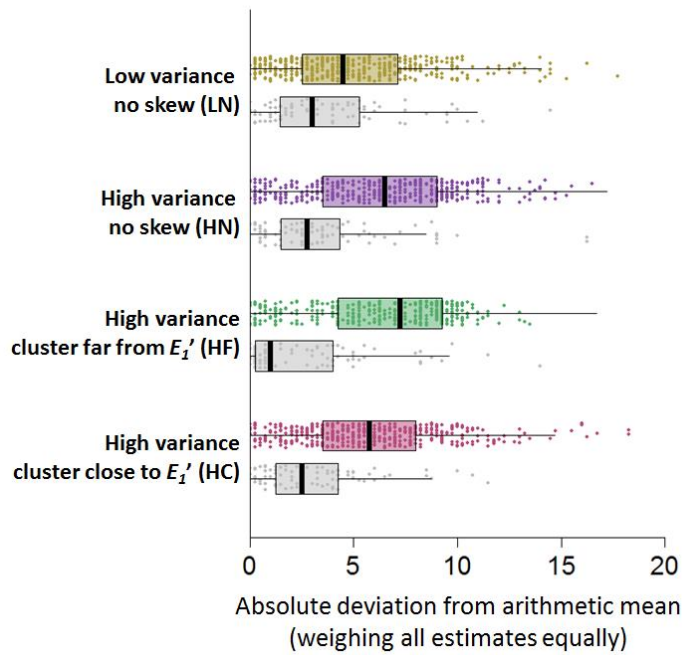


Fig. S3. Both in experimental (i.e., with observing the stimulus; colored dots and boxplots) and control (i.e., without stimulus; grey dots and boxplots) conditions, participants systematically deviated from an ‘ideal Bayesian observer’, who would weigh all estimates equally. Boxplots and dots show the absolute difference between participants’ second estimates and the arithmetic mean of all four estimates (see Methods more details). Participants deviated more from the arithmetic mean in the experimental conditions compared to the control conditions.

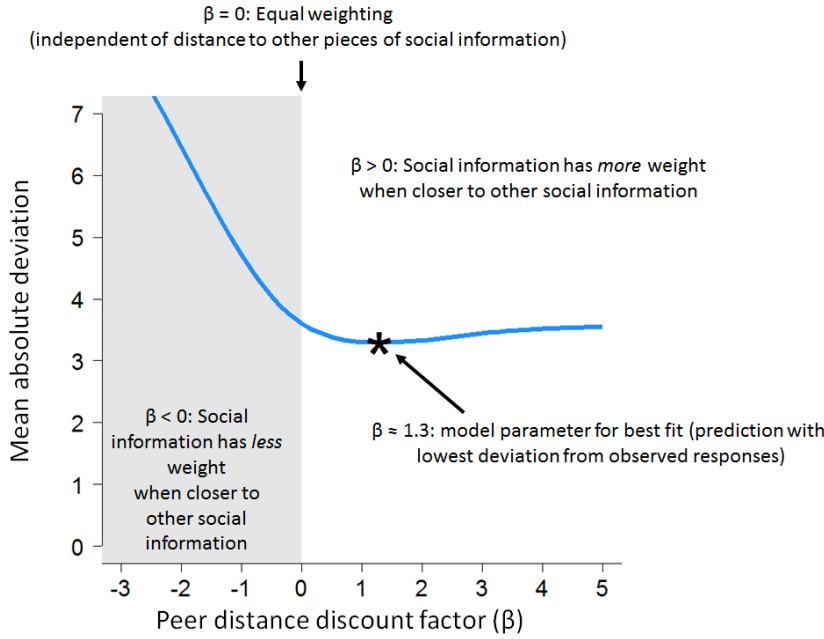


Fig. S4. Proximity-based weighting can account for responses (E_2') in the control rounds in which participants did not observe the stimulus. The blue line shows the mean absolute difference between per-round predictions and observed responses as a function of parameter β (capturing 'proximity weighting' as the extent of discounting of social information that is inconsistent with other social information). This model assumes that each response is an average of the pieces of social information (X_i), weighted according to their summed distance (d) to other pieces of social information. Formally, predicted responses are calculated as: $\widehat{E_2'} = \sum_{i=1}^4 w_i \cdot X_i$. In this formula, the weighting of each piece of social information (w_i) is determined by discount factor β : $w_i = \frac{d_i^{-\beta}}{\sum_{j=1}^4 d_j^{-\beta}}$, where $d_i = \sum_{j=1}^4 |X_i - X_j|$. When $\beta > 0$, more weight is assigned to social information when it is closer to other social information. We observe that the model predictions match the observed responses best when $\beta \approx 1.3$. We interpret this as evidence of proximity-based weighting: on average, participants tend to assign more weight to social information when it is closer to other social information.

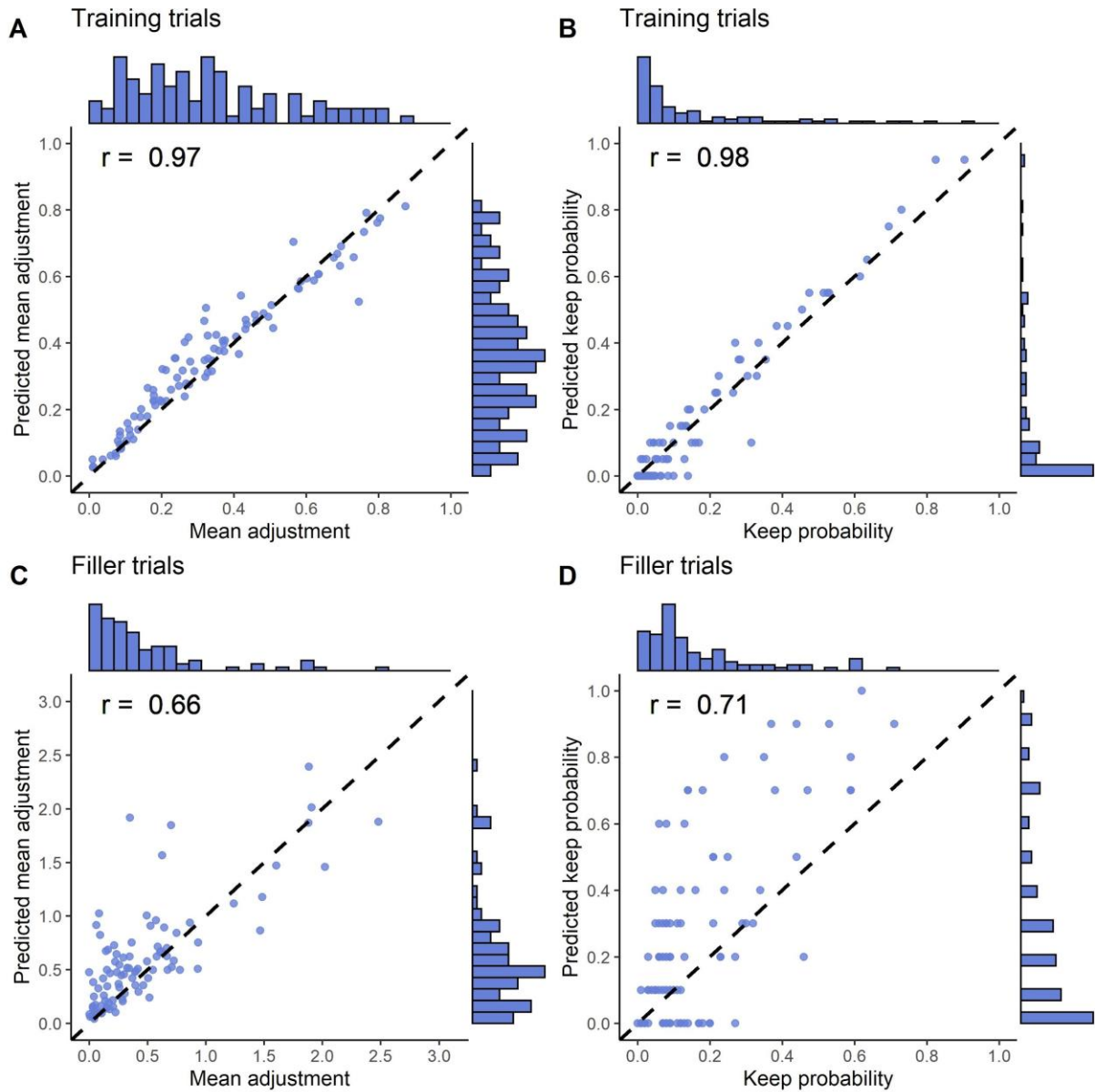


Fig S5. Correlation of predictions of the best-fitting cognitive model (y-axis) with the empirical data (x-axis). (A,B) Predictions for the 20 rounds of the 4 experimental conditions, to which the model was fitted (i.e. the ‘training set’). (A) Mean adjustment for each participant. (B) Fraction of rounds in which they chose to choose to ‘keep’ their first estimate. We observe that the predictions of the cognitive model closely match data in the experimental conditions. (C,D) Predictions for the ‘filler’ trials with social information randomly drawn from previous participants (on which the model was not trained). Also for these out-of-sample predictions, the model quite closely tracks the empirical data. As for previous predictions we sampled 10 estimates for each participant and trial. In each panel, numbers in the top left corner indicate Pearson’s correlations.

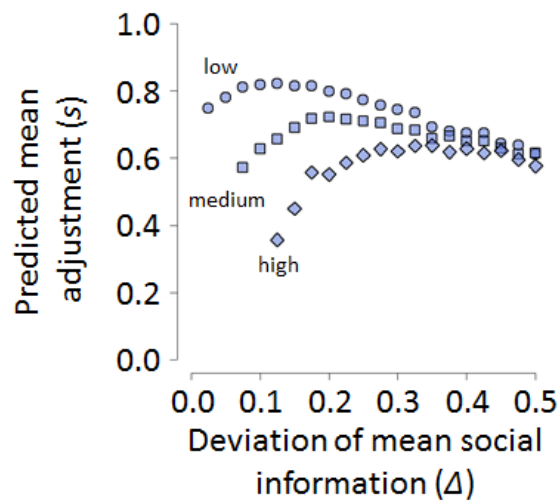


Figure S6. Social information tends to have the strongest impact when at intermediate distance. We simulated trials that systematically varied the distance between individuals' own first estimates and the mean social information ($\Delta = |\bar{X} - E_1| / E_1$). Dots show predicted mean adjustments as a function of the mean distance to three pieces of social information, for three levels of 'peer proximity', with peer estimates at 0, 3 or 6 numbers away from each other (respectively reflecting 'low', 'medium' or 'high' variance). We observe that mean shifts for each level are highest when the distance of social information is intermediate. This shows that our model is able to recover key results from previous studies showing this nonlinear effect of the distance of social information (34, 35). When the distance to mean social information is very low, individuals are very likely to keep their first estimates. When the distance to mean social information is very high, individuals are likely to compromise, but assign little weight to social information, again leading to reduced average adjustments. At intermediate distance, social information has the strongest impact: in those cases, the likelihood that individuals keep their first estimates is very close to zero, but when compromising, they still assign positive weight to social information. These effects hold across various levels of peer proximity. As expected, lower variance in social information leads to larger mean adjustments. Note that simulations with peers farther away from each other (i.e. 'high') start at higher values of Δ to avoid that the social information brackets the own first estimate. Including these 'bracketed' cases would show nonlinear adjustment patterns, because near $\Delta=0$ the distance to the nearest peer is a nonlinear function of Δ .

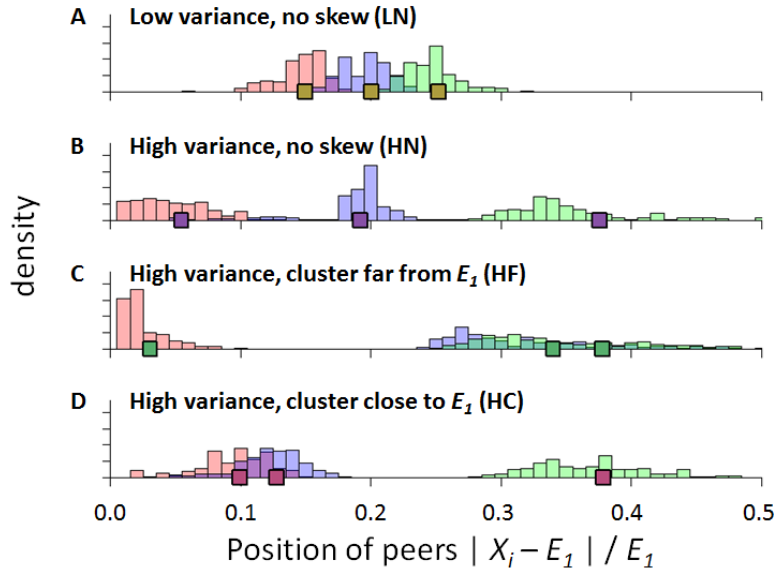


Fig S7. Relative position of peers as shown in each of the experimental conditions. Social information was based on data from a prerecorded pool of 100 MTurkers who completed the task without social information. As described in the Methods, we assigned an experimental condition to each of the 30 rounds and calculated for each value of E_1 the ‘triple’ of previous estimates that most closely matched the experimental condition (see Supplementary Methods for a formal description). The graph shows for each condition where the peers were located in the actual experiment, expressed as their absolute deviation from E_1 , divided by E_1 . Color coding: red: nearest peer, blue: middle peer, green: farthest peer. Colored squares show for each of the three peers their average position (cf. Fig. 1E).

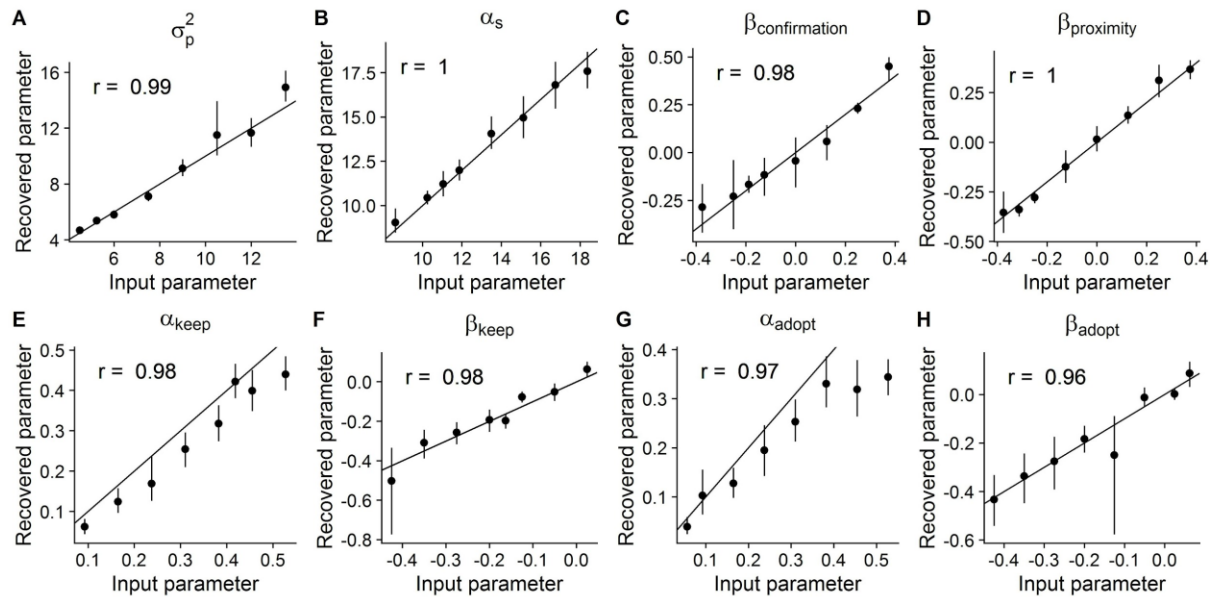


Fig S8. Recovery of all parameters of the full model. The dots and error bars indicate mean and the 95% CI of the posterior mean estimates. The diagonal line indicates the position expected under perfect parameter recovery. For the majority of recovered parameter estimates, the generating (input) parameters lie within the 95% CI. This indicates that the model parameters are identifiable. The parameter estimates for α'_{keep} and α'_{adopt} are lower than the generating parameter values. This is a result of choosing priors for the group-level parameters reflecting our expectation that high probabilities to keep or adopt are less likely (see Table S4). In each panel, numbers in the top left corner indicate Spearman's correlation coefficient.

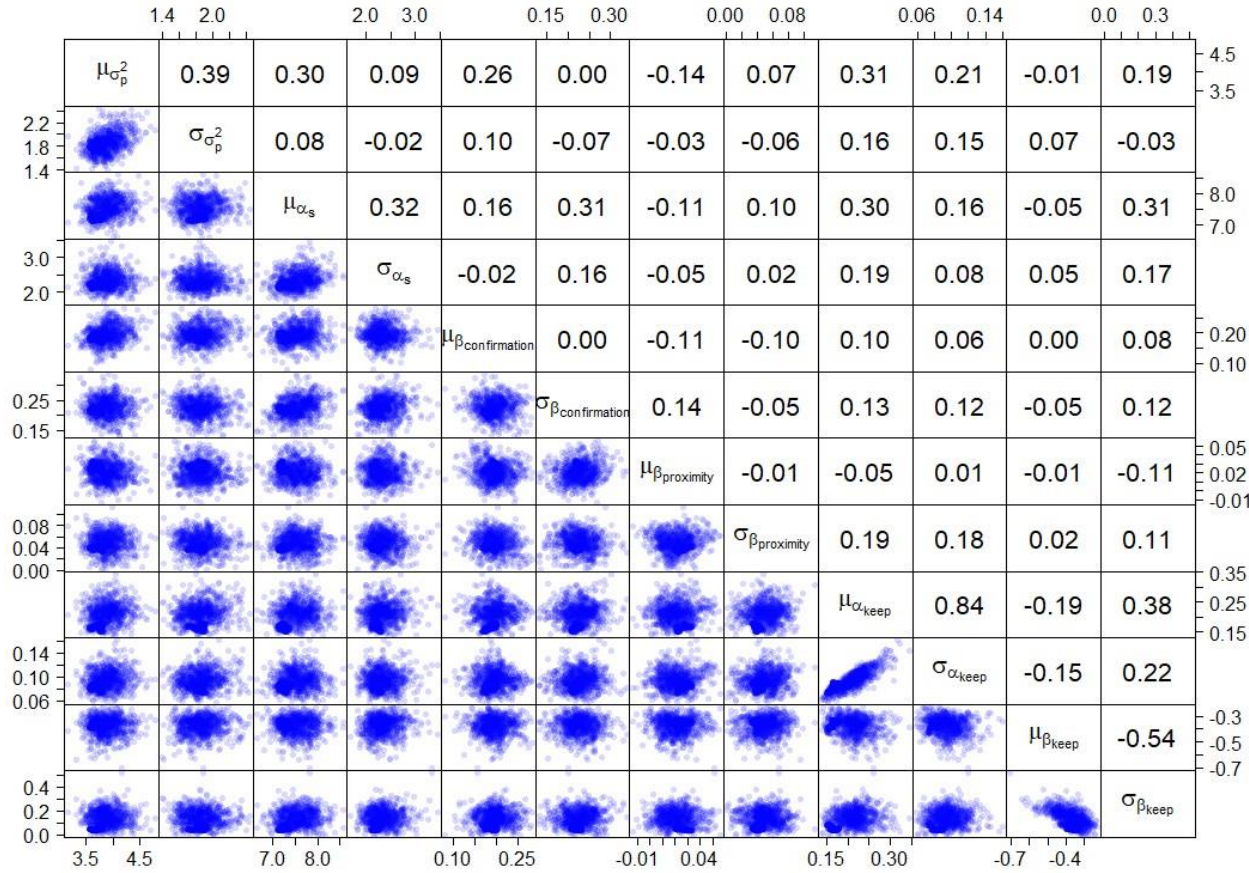


Fig. S9. The lower left side shows the joint posterior samples of the group-level means of the best fitting model. The upper right side displays Pearson correlation coefficient among these parameters. As expected we find correlations between some mean and corresponding variance parameters. Apart from this finding, we do not find very strong (non-linear) correlations among the parameters indicating that the parameters are relatively independent and, hence, no parameter combinations much more likely than others. To have comparable measures for all parameters we display the mean and variance of the group-level beta distribution of α'_{keep} via: $\mu = \frac{\alpha}{\alpha + \beta}$, $\tau = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$.

Supplementary Tables

Table S1. Determinants of participants' social information use. Numbers on the left hand side show results from a Bayesian linear mixed model (fitted with the R-package *brms* (69) using default priors) fitted to participants' average adjustments towards the mean social information across the rounds of each of the experimental conditions, with 'participant' as varying intercept. Numbers between brackets indicate 95% posterior credibility intervals. On the right hand side, we show pairwise comparisons between experimental conditions. Average adjustments were calculated as the relative distance adjusted towards the mean social information (Fig. 1D), across the five rounds of each of the four experimental conditions (LN, HN, HF, HC; Fig. 1E), yielding four data points for each participant. We omitted those rounds in which a participant adjusted away from the mean social information. The LN condition was used as the baseline. Relative to that baseline, the HN and HF conditions had a strong negative effect. The HC condition had a smaller negative effect. Pairwise comparisons indicate that, except for the HN and HF conditions, all pairs of experimental conditions differed significantly from each other. Age and gender did not seem to affect social information use. To quantify the participants' constancy of displayed behaviour across conditions we also derived the commonly used index of 'repeatability' for this model ($R=0.790$ [0.697, 0.839]) (69).

	Average adjustment		
	Estimate [95% CI]	Pairwise comparison	Estimate [95% CI]
Low variance, No skew (LN); baseline	0.39 [0.20, 0.57]	HN - LN	-0.13 [-0.15, -0.10]
High variance, No skew (HN)	-0.13 [-0.16, -0.09]	HF - LN	-0.14 [-0.16, -0.11]
High variance, cluster far from E1 (HF)	-0.14 [-0.17, -0.10]	HC - LN	-0.05 [-0.08, -0.02]
High variance, cluster close to E1 (HC)	-0.05 [-0.08, -0.02]	HF - HN	-0.01 [-0.03, 0.02]
Age	0.00 [-0.00, 0.00]	HC - HN	0.08 [0.05, 0.10]
Gender (0=male, 1=female)	0.02 [-0.07, 0.12]	HC - HF	0.09 [0.06, 0.11]
N	95		
n	380		

Table S2. Marginal effects of experimental conditions on participants' use of adjustment strategies. Values indicate for each condition the predicted relative frequencies of adjustment strategies - keeping, compromising, adopt the nearest peer (X_1), and all 'other' cases pooled - from a mixed multinomial regression with 'participant' as varying intercept (cf. insets Fig. 2A-D of the main text), fitted with the R-package 'brms' (70) using default priors. Values in brackets indicate 95% posterior credibility intervals. Apart from the experimental conditions, the regression model also included age and gender, neither of which had credible non-zero effects on any of the relative frequencies.

condition	keep	compromise	copy X_1	other
LN	0.07 [0.04, 0.10]	0.79 [0.70, 0.86]	0.02 [0.01, 0.05]	0.12 [0.07, 0.19]
HN	0.16 [0.11, 0.22]	0.26 [0.18, 0.37]	0.07 [0.04, 0.11]	0.51 [0.37, 0.63]
HF	0.16 [0.11, 0.23]	0.08 [0.05, 0.13]	0.09 [0.05, 0.15]	0.67 [0.54, 0.77]
HC	0.06 [0.04, 0.09]	0.67 [0.56, 0.76]	0.06 [0.03, 0.10]	0.21 [0.13, 0.32]

Table S3. The comparison of all versions of the cognitive models with all possible combinations of considered features. We compared models with (1) and without (0) each of the four features (keeping, adopting, confirmation-based and proximity-based weighting) and calculated the looic. The differences in expected log pointwise predictive density ('elpd_diff') indicate the expected predictive accuracy of each model compared to the model with the lowest looic (i.e., rank 1), with 'se_diff' indicating the corresponding standard error. The best-fitting model includes the keep heuristic, proximity and confirmation, but not the 'adopt' heuristic. Note that the model including 'adopt' (i.e., rank 2) has only marginally lower predictive accuracy. Inspection of the results from the full model (including all features) reveals that predicted probabilities of adopting are very low.

Keep	Adopt	Proximity	Confirmation	Looic	elpd_diff	se_diff	Rank
1	0	1	1	8222	0	0	1
1	1	1	1	8225	-1.55	3.62	2
1	0	0	1	8237	-7.66	5.52	3
1	1	0	1	8247	-12.61	6.37	4
1	1	1	0	8433	-105.62	16.71	5
1	0	1	0	8444	-111.27	16.25	6
1	1	0	0	8479	-128.57	17.12	7
1	0	0	0	8498	-137.92	16.78	8
0	0	1	1	8713	-245.72	26.67	9
0	0	0	1	8719	-248.75	26.62	10
0	1	1	1	8720	-249.18	26.97	11
0	1	0	1	8726	-251.91	26.75	12
0	1	1	0	8982	-379.81	30.67	13
0	0	1	0	8995	-386.8	31.12	14
0	1	0	0	9014	-396.15	31.35	15
0	0	0	0	9037	-407.4	31.02	16

Table S4. The parameters shaping the group-level distributions, their priors, their upper (lower) bounds and the model estimates of the best fitting model (i.e, without the ‘adopt’ heuristic). The parameters μ and τ describe the mean and standard deviation of group-level normal distributions and α and β the two parameters shaping the group-level beta distributions. Standard deviations and parameters describing the beta distributions were truncated at 0.01 to avoid zero or negative values. Subscripts p and s respectively indicate personal and social information; the other subscripts refer to their respective model feature (confirmation or proximity weighting; heuristics of keeping or adopting). The right hand side column shows the parameter estimates of the best-fitting model, with brackets indicating 95% posterior credibility intervals. This model did not include the ‘adopt’ heuristic (see Table S3) so no values are shown for the parameters associated with that heuristic.

Group-level parameters	Priors	Bounds (max/min)	Parameter estimates
$\mu_{\sigma_p^2}$	N(10, 5)	1.0 / Inf	3.91 [3.50, 4.40]
$\tau_{\sigma_p^2}$	N(0, 1)	0.01 / Inf	1.87 [1.54, 2.30]
μ_{α_s}	N(10, 5)	1.0 / Inf	7.53 [6.91, 8.20]
τ_{α_s}	N(0, 1)	0.01 / Inf	2.35 [1.92, 2.87]
$\mu_{\beta_{confirmation}}$	N(0, 0.2)	-0.5 / 0.5	0.20 [0.13, 0.26]
$\tau_{\beta_{confirmation}}$	N(0, 0.2)	0.01 / Inf	0.22 [0.15, 0.29]
$\mu_{\beta_{proximity}}$	N(0, 0.2)	-0.5 / 0.5	0.02 [0.00, 0.04]
$\tau_{\beta_{proximity}}$	N(0, 0.2)	0.01 / Inf	0.05 [0.01, 0.08]
$\alpha_{\alpha'_{keep}}$	N(1, 0.4)	0.01 / Inf	0.18 [0.12, 0.30]
$\beta_{\alpha'_{keep}}$	N(5, 2)	0.01 / Inf	0.71 [0.45, 1.11]
$\mu_{\beta_{keep}}$	N(0, 0.5)	-Inf / Inf	-0.36 [-0.52, -0.25]
$\tau_{\beta_{keep}}$	N(0, 1)	0.01 / Inf	0.14 [0.02, 0.30]
$\alpha_{\alpha'_{adopt}}$	N(1, 0.4)	0.01 / Inf	
$\beta_{\alpha'_{adopt}}$	N(5, 2)	0.01 / Inf	
$\mu_{\beta_{adopt}}$	N(0, 0.5)	-Inf / Inf	
$\tau_{\beta_{adopt}}$	N(0, 1)	0.01 / Inf	

Table S5. Pearson correlations for the mean posterior parameter estimates across participants of the best-fitting model, with 95% posterior credibility intervals in brackets. Overall, we did not find strong correlations between pairs of parameter estimates, with three exceptions: (i) Participants who put less weight on themselves (higher σ_p^2) tend to apply stronger proximity weighting (higher $\beta_{proximity}$) and (ii) tend to have a lower tendency to keep their first estimates (lower α'_{keep}). (iii) The keep heuristic of participants with a stronger tendency to keep first estimates (higher α'_{keep}) is less sensitive to the distance of the nearest peer (less negative β_{keep}).

	σ_p^2	α_s	$\beta_{confirmation}$	$\beta_{proximity}$	α_{keep}	β_{keep}
σ_p^2	1.00					
α_s	-0.02 [-0.22, 0.18]	1.00				
$\beta_{confirmation}$	-0.13 [-0.32, 0.07]	0.00 [-0.20, 0.20]	1.00			
$\beta_{proximity}$	0.37 [0.18, 0.53]	-0.13 [-0.32, 0.08]	-0.10 [-0.30, 0.10]	1.00		
α_{keep}	-0.21 [-0.40, -0.01]	0.02 [-0.19, 0.22]	0.12 [-0.09, 0.31]	0.00 [-0.20, 0.20]	1.00	
β_{keep}	-0.01 [-0.21, 0.19]	-0.01 [-0.21, 0.19]	0.02 [-0.19, 0.22]	0.00 [-0.20, 0.20]	0.37 [0.18, 0.53]	1.00

Table S6. Description of the model parameters and the parameters of their group-level distribution.

Model feature	Parameter	Group-level distributions	Description
Keep			
Keep intercept	α'_{keep}	$Beta(\alpha_{\alpha'_{keep}}, \beta_{\alpha'_{keep}})$	A value between zero and one describing the baseline probability of keeping the first estimate.
Keep slope	β_{keep}	$N(\mu_{\beta_{keep}}, \tau_{\beta_{keep}})$	The influence of the distance of the nearest peer on the probability to keep the first estimate.
Adopt			
Adopt intercept	α'_{adopt}	$Beta(\alpha_{\alpha'_{adopt}}, \beta_{\alpha'_{adopt}})$	A value between zero and one describing the baseline probability of adopting the estimate of a peer.
Adopt slope	β_{adopt}	$N(\mu_{\beta_{adopt}}, \tau_{\beta_{adopt}})$	The influence of distance on the probability to adopt the estimate of a peer.
Compromise			
Uncertainty first estimate	σ^2_p	$N(\mu_{\sigma^2_p}, \tau_{\sigma^2_p})$	Uncertainty associated with personal information (i.e. the first estimate, E_1)
Intercept uncertainty peer estimate	α_s	$N(\mu_{\alpha_s}, \tau_{\alpha_s})$	The intercept uncertainty associated with the peer estimates (X_i)
Confirmation-based weighting	$\beta_{confirmation}$	$N(\mu_{\beta_{confirmation}}, \tau_{\beta_{confirmation}})$	The influence of closeness of the peer estimate to the first estimate on the uncertainty associated with the peer estimate
Proximity-based weighting	$\beta_{proximity}$	$N(\mu_{\beta_{proximity}}, \tau_{\beta_{proximity}})$	The influence of proximity of the peer estimate to other peers on the uncertainty associated with the peer estimate

Supplementary Methods

Definition of experimental conditions. Participants faced four experimental conditions in which they could adjust their first estimates based on three pieces of social information. These conditions varied in the variance and skewness of the social information (Fig. 1E, main text). For each round, for each possible first estimate (E_1) we considered each possible triple of unique prerecorded estimates, and calculated the first three moments of its distribution (mean μ , variance σ^2 and skewness γ). To determine which triple would be shown in a given condition in a given round for a given value of E_1 , we used a cost function that assigned penalties (L) to deviations from the target mean (T_μ), target variance (T_{σ^2}) and target skewness (γ). For each round, for each possible value of E_1 we selected the triple with the lowest L . The cost functions L for each condition are given in the below table, which shows the penalties for deviations from the target mean, variance and skewness in separate columns:

Mean		Variance		Skewness
Low variance, no skew (LN)				
$100 \cdot \mu - T_\mu $	+	$10 \cdot \sigma^2 - T_{\sigma^2} $	+	$10 \cdot \gamma $
High variance, no skew (HN)				
$100 \cdot \mu - T_\mu $	+	$10 \cdot \sigma^2 - T_{\sigma^2} $	+	$500 \cdot \gamma $
High variance, cluster far from E_1 (HF)				
$100 \cdot \mu - T_\mu $	+	$10 \cdot \sigma^2 - T_{\sigma^2} $	-	$1,000 \cdot \gamma'$
High variance, cluster close to E_1 (HC)				
$100 \cdot \mu - T_\mu $	+	$10 \cdot \sigma^2 - T_{\sigma^2} $	+	$1,000 \cdot \gamma'$

In all conditions, T_μ was set to deviate 20% from E_1 . We held this distance fixed to avoid possible effects of the mean deviation of social information on its impact on behavior ((34, 35); see also Fig. S6). For further standardization, T_μ was always in the direction of the true value A (Yaniv 2000; Molleman et al. 2019). Formally, $T_\mu = 1.2 \cdot E_1$ if ($E_1 > A$) and $T_\mu = 0.8 \cdot E_1$ if ($E_1 < A$). We set $T_{\sigma^2} = 10$ for the LN condition and $T_{\sigma^2} = 100$ for all other experimental conditions. For the LN and HN we aimed for symmetric distributions so we penalized positive absolute values of γ . For HF and HC, target skewness depended

on whether E_1 was higher or lower than A. To this end, we used γ' which was equal to γ if $E_1 > A$ and equal to $-\gamma$ if $E_1 < A$. This procedure ensured that participants faced well-defined experimental conditions (Fig. S7). For the 10 'filler' rounds, which were intermixed with the experimental rounds, we randomly drew social information from the prerecorded pool.

We further implemented two control conditions completed in separate blocks of the experimental session (these blocks were completed in randomized order). First, participants completed trials in which they did not observe the stimulus themselves, but only observed the estimates of four peers (Fig. S2). The distribution of these peer estimates emulated the distributions of social information in each of the experimental conditions, enabling us to compare how individuals integrate personal and social information with a control in which individuals integrate four pieces of information, none of which is their first estimate. Second, participants could observe the estimate of only one peer whose deviation from the individuals' first estimate matched that of the mean deviation in the four experimental conditions. The results from this one-peer control condition are not the focus of this paper and will not be reported here.

Cognitive model. To analyse how individuals integrated social information we developed a mixture model approach, where individuals first select their adjustment strategy ('keep' their own personal estimate, 'adopt' the estimate of the nearest peer, or 'compromise' towards the social information). For 'keep' and 'adopt', we assume that these two strategies are chosen with mixture probabilities $P(\text{keep})$ and $P(\text{adopt})$. Compromising is chosen with the complementary probability $1 - P(\text{keep}) - P(\text{adopt})$.

Implementation of compromising. We model compromising as a process of Bayesian updating. The individual has two sources of information to infer the number of animals (N): (i) The individual's personal belief about the number of animals (E_p) and social information provided by the peers (X_s), with $s=1:k$, and k being the group size. We model the personal belief as a discretized normal probability distribution centered around first estimate (E_1) and associated with uncertainty (σ_p^2):

$$p(E_p | N) \sim \text{Norm}(E_1, \sigma_p^2).$$

Similarly, we model social information (SI_s) as a discretized normal probability distributions centered around peer estimates X_s , with associated uncertainties (σ_s^2):

$$p(SI_s | N) \sim \text{Norm}(X_s, \sigma_s^2).$$

With only one peer (i.e. $k = 1$) we obtain the (posterior) probability of N animals by applying Bayes' rule:

$$p(N | E_p, SI_1) = \frac{p(E_p | N) * p(SI_1 | N) * p(N)}{p(E_p, SI_1 | N)}$$

Here, $p(N | E_p, SI_1)$ is the updated 'posterior' belief about the number of animals (N) given personal and social information (E_p and SI_1 , respectively). We assume that before observing the images with the animals, individuals have no prior expectations regarding N . We therefore model $p(N)$ as a uniform distribution.

When individuals observe three peers (as in our experimental conditions), a similar updating procedure is conducted with all peer estimates:

$$p(N | E_p, SI_1, SI_2, SI_3) = \frac{p(E_p | N) * p(SI_1 | N) * p(SI_2 | N) * p(SI_3 | N) * p(N)}{p(E_p, SI_1, SI_2, SI_3 | N)}$$

Note that the order of social information is not affecting the outcome of this updating process.

Based on our empirical results and results from the literature, we distinguish between two features of how individuals weigh social information, capturing possible effects of 'confirmation' and 'proximity' (see main text). In our implementation, this weighting is reflected in the uncertainty associated with a peer estimate (σ_s^2), with higher values of uncertainty indicating less weight assigned to an estimate. In the probability density functions, we set the minimum values to 10^{-30} to avoid outcome probabilities of zero.

Model fitting. We fitted the model using a hierarchical Bayesian inference technique implemented with *RStan* in *R* (67, 68). We used a hierarchical model structure, in which each model parameter had a higher-order prior (see Table S6). For all model parameters (except two, see below), priors were normal distributions with hyperparameters describing the mean and variance. The only exceptions to this were α_{keep} and α_{adopt} . Many participants (almost) never chose to keep their first estimate or to adopt the estimate of a peer (cf. Fig. 2F), rendering a zero-inflated distribution for these two parameters. To account for individual differences in these two parameters, we used more flexible (non-normal) group-level distributions. In particular, we implemented α_{keep} and α_{adopt} (continuous numbers) via a transformed parameter $\alpha_{keep} = \text{logit}(\alpha'_{keep})$ or $\alpha_{adopt} = \text{logit}(\alpha'_{adopt})$. This

restricts parameters α'_{keep} and α'_{adopt} between zero and one, and allows us to describe these parameters with beta distributions, each having two hyperparameters controlling their shape (see Table S6). Further, note that for the fitting process, we centered the predictors d_s and τ_s to their mean values.

In the fitting procedure we ran 4 chains in parallel with 2,000 iterations each and discarded the first 1000 as burn-in. We reduced the memory load by thinning the chains with a factor of 5. We investigated the importance of four model features: (i) keep heuristic; (ii) the adopt heuristic; and when compromising, (iii) confirmation-based weighting (assigning weight to social information depending on its distance of the personal first estimate), (iv) proximity-based weighting (assigning weight to social information depending its distance to other peer estimates). The importance of each of these features was evaluated by calculating the leave-one-out cross-validation (looic; (55)) of the models containing all possible combinations of these features (16 in total; Table S3). We quantified the importance of a feature by calculating the average reduction of the looic when the feature was included rather than excluded (Fig. 2B). We report the fittings of the model with the lowest looic (Table S4). Visual inspection of Markov chains and the Gelman Rubin statistic $\hat{R}$ indicated that all Markov chains of all of the 16 investigated models converged.

Parameter recovery analysis. To ensure that all model parameters are identifiable and influence judgments in an interpretable manner, we conducted a parameter recovery analysis. Specifically, we drew eight sets of model parameters with a quasi-random number generator using the sobol sequence (from the *R* package *randtoolbox* (71)) to ensure a uniformly covered multidimensional parameter space within a reasonable parameter range with following ranges: σ^2_p varies within [3,15], α_s within [7,20], $\beta_{confirmation}$ within [-0.5,0.5], $\beta_{proximity}$ within [-0.5,0.5], α'_{keep} within [0,0.6], β_{keep} within [-0.5,0.1], α'_{adopt} within [0,0.6], β_{adopt} within [-0.5,0.1].

For each parameter set, we generated second estimates based on first estimates and observed social information from our four experimental conditions. We also used the same set of stimuli and the same number of individuals who participated in our experiment. This allows us to test whether the hierarchical implementation and sample size allow for an accurate identifiability of model parameters. The generated dataset thus consists of 100 participants with 20 synthetic second estimates each. We then fitted the model for each parameter set using the identical analysis as described in the main text. We report the mean and the 95% CI of the group-level parameter estimate. To compare input parameters and estimates of α_{keep} and α_{adopt} we used the mean of the group-level beta distribution

$\mu = \frac{\alpha}{\alpha + \beta}$. We measured the relationship of input and recovered parameters by calculating Spearman's correlation coefficient. For all parameters we find a strong positive correlation between input and recovered parameters indicating that the empirically found parameters describe distinguishable aspects of judgements and are interpretable in their magnitude (Fig. S9).

Experimental materials

Participants were recruited from the crowdsourcing platform Amazon Mechanical Turk (MTurk). On that platform, our experiment was advertised as a 'Human Intelligence Task' (HIT) with a link that led to the experimental screens. Upon completion of the experiment, participants received a code that they could fill out on MTurk to receive their participation fee of \$4.50, plus a performance-specific bonus.

The experimental session consisted of 3 blocks (referred to as 'Task I, II and III' in the instructions for participants), the order of which was randomized across participants.

Block 1: Participants observed an image and made their first estimate of the number of animals on it. Then, they could observe the estimate of *three* participants who completed the task before, and then make a second estimate. Participants completed five rounds for each of the experimental conditions (cf. Fig. 1E of the main text), plus 10 'filler rounds' in which they observed three randomly drawn previous participants. So, in this block, participants completed 30 rounds in total; the participants' responses in the experimental rounds of this block are the main focus of this study

Block 2: Like in Block 1, participants observed an image and made their first estimate of the number of animals on it. Then, they could observe the estimate of *one* participant who completed the task before, and then make a second estimate. This was repeated for five rounds (with a new image showing another species of animal in every round). NB: the data for this control condition is not the focus of this paper, and therefore its results are not presented here.

Block 3: Participants did *not* observe an image, but had to make an estimate based on the estimates of four participants who completed the task before. The five rounds of this task mimicked the four experimental conditions plus a random round (see Methods).

Below we show screenshots of the experiment. We add notes in between screenshots where appropriate, for clarification and to prevent repetitiveness.

Welcome

In this HIT we ask you to complete **three tasks** (Task I, Task II and Task III).
During these tasks you can earn points. The number of points you earn will depend on your responses.
In each task, you can earn up to **100 points**.

At the end of this HIT, your points from Task I, Task II and Task III will be added up and converted into your bonus (**100 points = \$1.00**).

Your bonus will be paid on top of your guaranteed participation fee of **\$4.50**.
Upon completion, you will receive a random code to collect your payment on MTurk.

In total, this HIT will take about 30-40 minutes. Please complete it without interruptions.
We finalize this HIT with a brief questionnaire.

Click below to proceed to the task instructions.
Please read these instructions carefully.

Continue

HIT description

In each of the tasks in this HIT you have to make **estimates**.
The number of points you earn depends on **how accurate** your estimates are.

Please click below to continue to Task I.

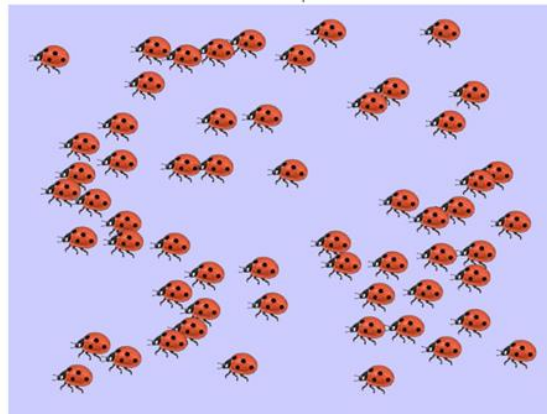
Continue

Task I: instructions 1 of 3

This task consists of 30 rounds.

At the beginning of each round, you will observe an image showing a number of animals.

For example:



The image will disappear after 6 seconds, upon which you have to estimate how many animals were displayed.
The more accurate your estimate, the more points you can earn. We explain this later.

Continue

Task I: instructions 2 of 3

Once the image has disappeared, you have to enter your estimate of how many animals were displayed, by using a slider.
This is your estimate for **part A** of a round.

Once you have entered your estimate, **part B** of the round begins.
You can observe the *part A* estimate of three other MTurkers.

Over 100 MTurkers participated in a previous session in which they completed this task.
In each round, you can observe the *part A* estimate of **three** of these previous MTurkers.

The previous MTurkers saw the same image as you. They also saw it for **6 seconds**.
After the image disappeared, they also had to **estimate** how many animals were displayed.
They could also earn a higher bonus if their estimate was more accurate.

You then have to **enter a second estimate**.
You can enter the **same** estimate as in part A, or adjust it as you wish.
This is your estimate for **part B** of a round.

*Note: please enter your estimates within the time limit on your screen.
If you do not make your estimate and press 'Continue' before the timer reaches zero,
you will be removed from the HIT and we will not be able to pay you!*

Once you have entered your part B estimate, the round is over and a new round begins.

Continue

Go back

Task I: instructions 3 of 3

Your bonus earnings

The more accurate your estimates, the more points you can earn in this task.

At the end of this HIT, the points you earn are converted into your bonus earnings.

Your bonus for this task is calculated as follows.

Once you have completed this HIT, the computer will randomly select 1 of the 30 rounds.

If you estimated the number of animals *exactly right*, you earn 100 points.

For each number that you are off, we subtract 5 points.

The number of points you earn cannot become negative.

For example, if the actual number of animals in the image was 60, and your estimation was 53, you were 7 off.
This would mean that we subtract $7 \times 5 = 35$ points. Your earnings for that estimate would be $100 - 35 = 65$ points.

Click 'Continue' if you understood your task.
A brief quiz will follow to check your understanding.

Continue

Go back

Check for understanding

To check your understanding of the task, please indicate for each of these statements whether they are correct or incorrect.

In each round of this task you will view an image. You have to estimate how many animals were displayed in it.

☒ correct ☐ incorrect

Once you have entered your estimate, you can observe the estimate of three other MTurkers who completed this task before. You can then make a second estimate.

☒ correct ☐ incorrect

The more accurate your estimates, the more points you can earn.

☒ correct ☐ incorrect

Continue

Back to instructions

Participants had to answer the check for understanding correctly before proceeding to the task.

Round 1

You have answered all control questions correctly! The task starts now.

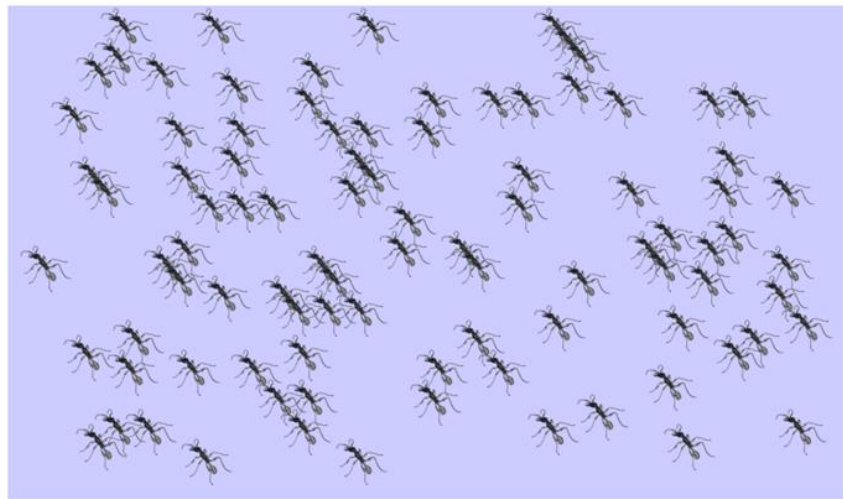
When you click below, an image will appear showing a number of ants.
After 6 seconds, the image disappears.

You have to estimate how many ants there were, using a slider.

You have to enter your estimate **within the time limit**.

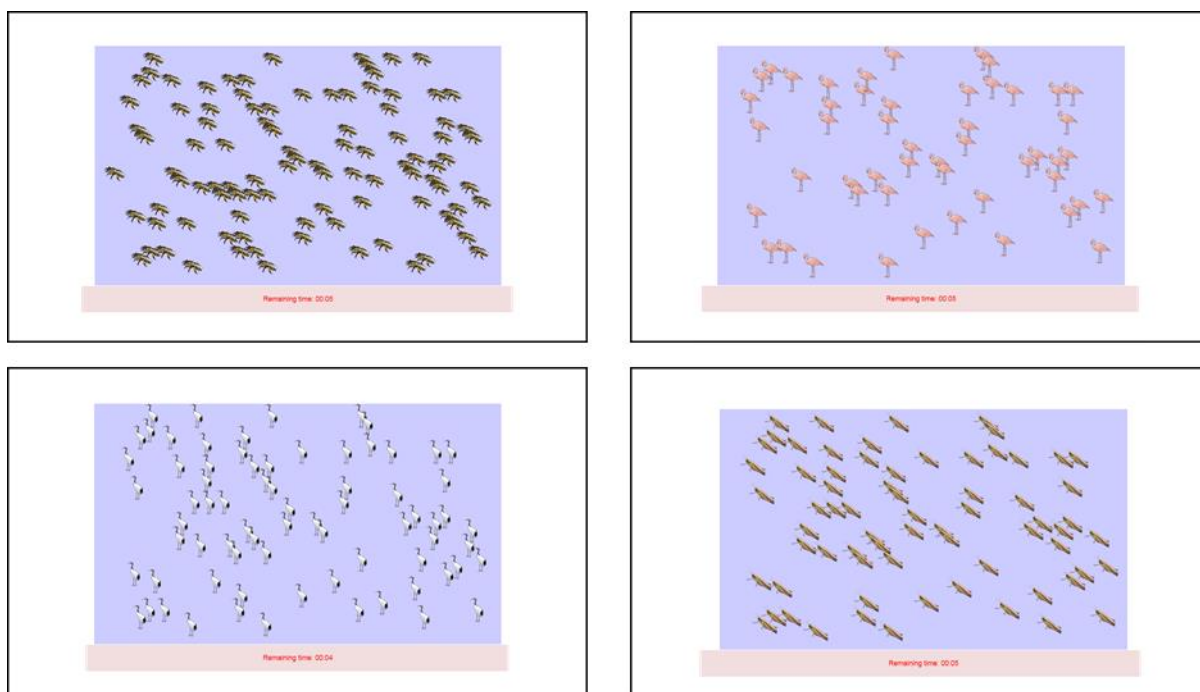
Click below when you are ready.

Continue



Remaining time: 00:05

Participants could view the images for 6 seconds. We used five species of animal on the images. Across rounds, the sequence of shown species was fixed: ants-bees-flamingos-cranes-cricket-ants-bees-flamingos-cranes-cricket-ants-etc.



Round 1: part A

How many ants were shown in the picture?

Please use the slider to indicate your estimate.

Your estimate is: 66

Continue

Remaining time: 00:23

This is the decision screen for entering the *first estimate*. Participants used the slider to indicate their estimate. The slider was limited at 1 and 150 (NB: the number of animals shown in the images ranged from 50-100), and initialized at the left hand side without showing a running number. While moving the slider, the running number was shown in blue. Participants could not proceed to the next screen without moving the slider.

We now show the screens in which participants entered their second estimates. In these screens, participants observed social information in the form of the estimates of three others who completed the task before. As described in Fig. 1D of the main text and in the Methods, we manipulated social information such that the conditions differed in the variance and the skewness of the distribution of social information, while holding fixed the mean deviation from a participant's first estimate (to the extent that the prerecorded pool of information allowed; see Methods for details and Fig. S7 for distributions of the three peers in each condition). The order of the rounds of the experimental conditions were randomized, and were intermixed with 'filler' rounds in which social information was randomly drawn from the prerecorded pool of estimates.

The following five screenshots show an example of this screen for each of the experimental conditions (plus the random condition for the 'filler' rounds). For the sake of exposition, we have filled out '60' as the first estimate in each of these rounds when making the screenshots.

Round 21: part B

The blue square shows your part A estimate.
The red dots show the estimates of three previous MTurkers.

Please use the slider to indicate your second estimate.

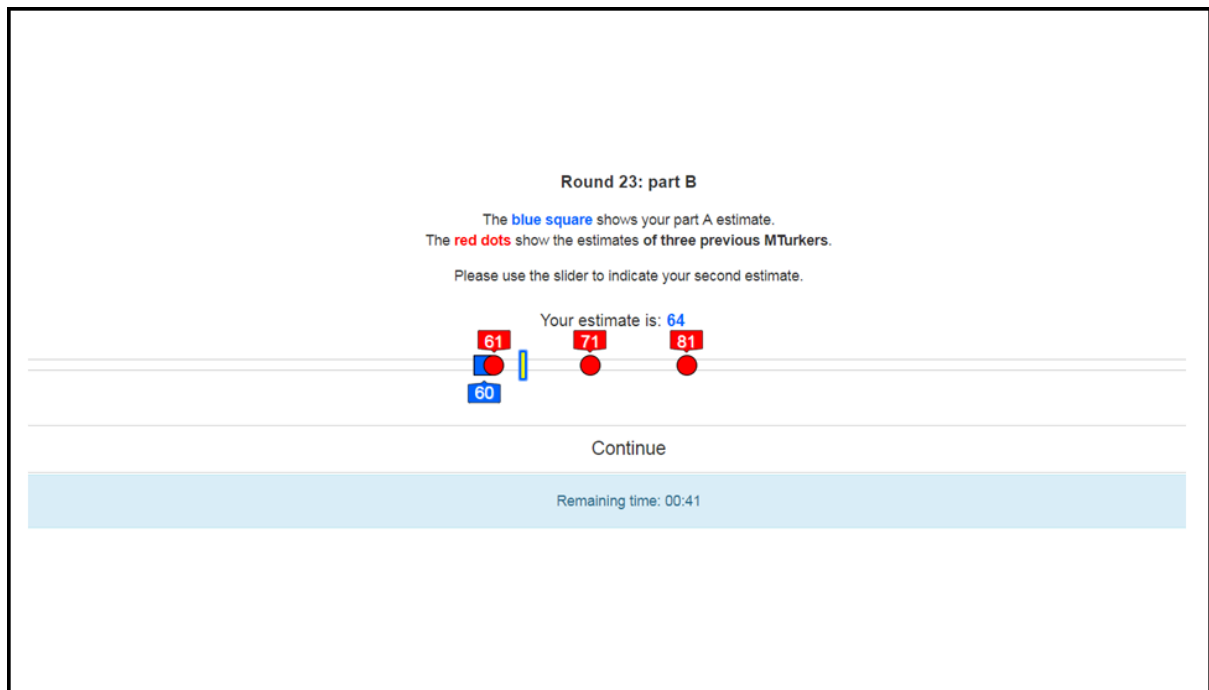
Your estimate is: 65

60 69 72 75

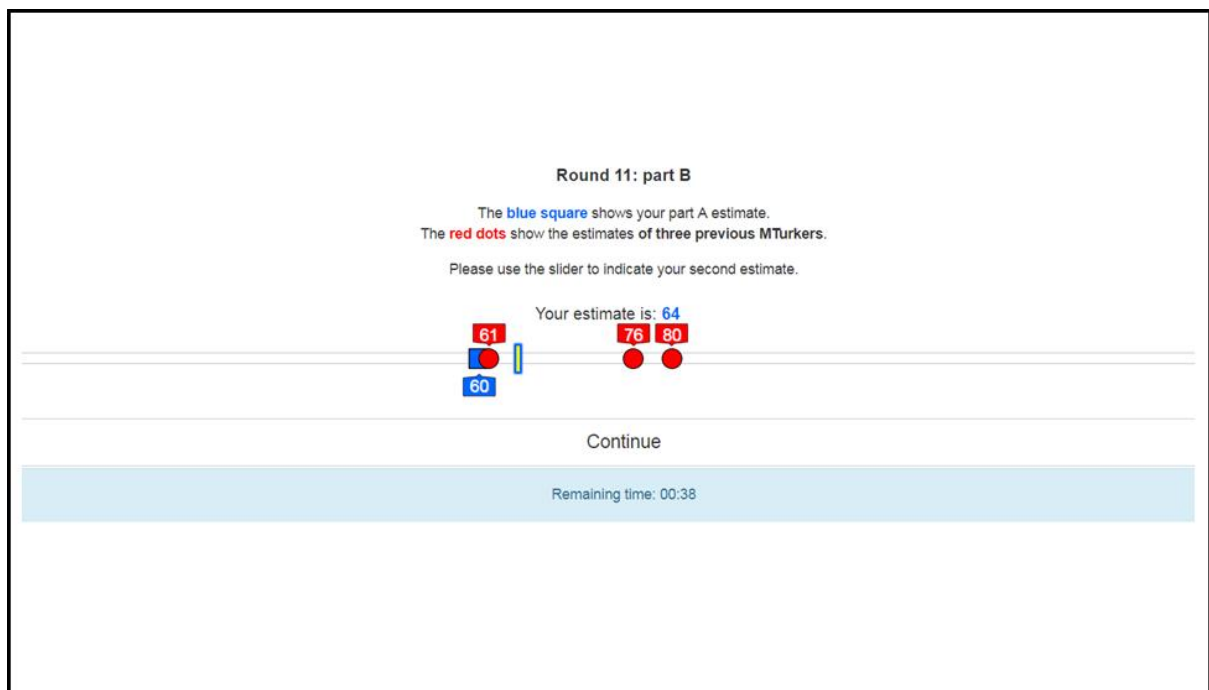
Continue

Remaining time: 00:39

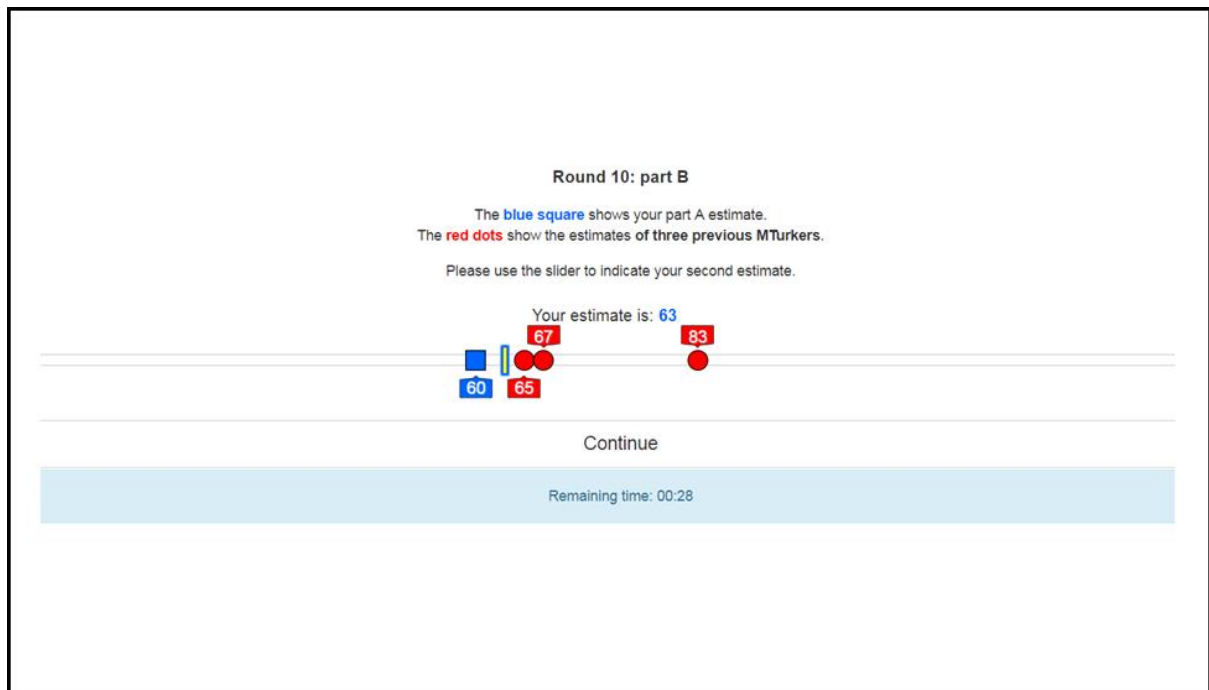
Low variance, no skewness (LN); peers strongly agreed with each other.



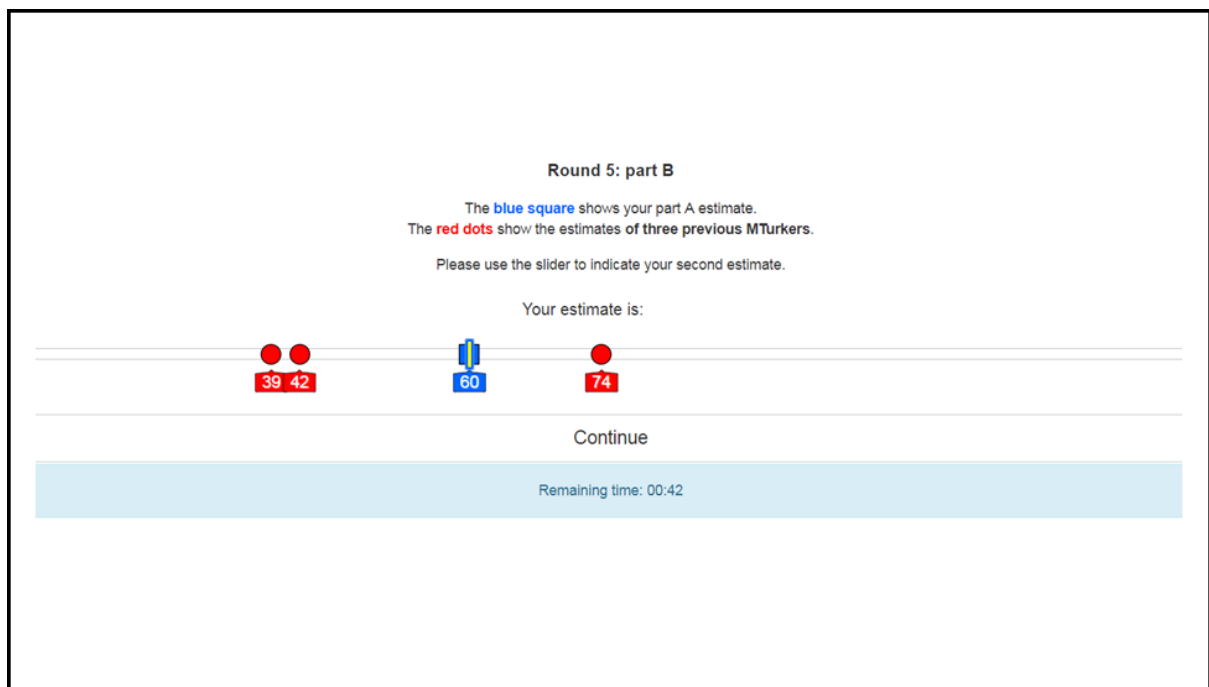
High variance, no skewness (HN). Peers disagreed amongst each other.



High variance, cluster far from E1 (HF). One peer strongly agreed with the participant, while two other strongly disagreed.

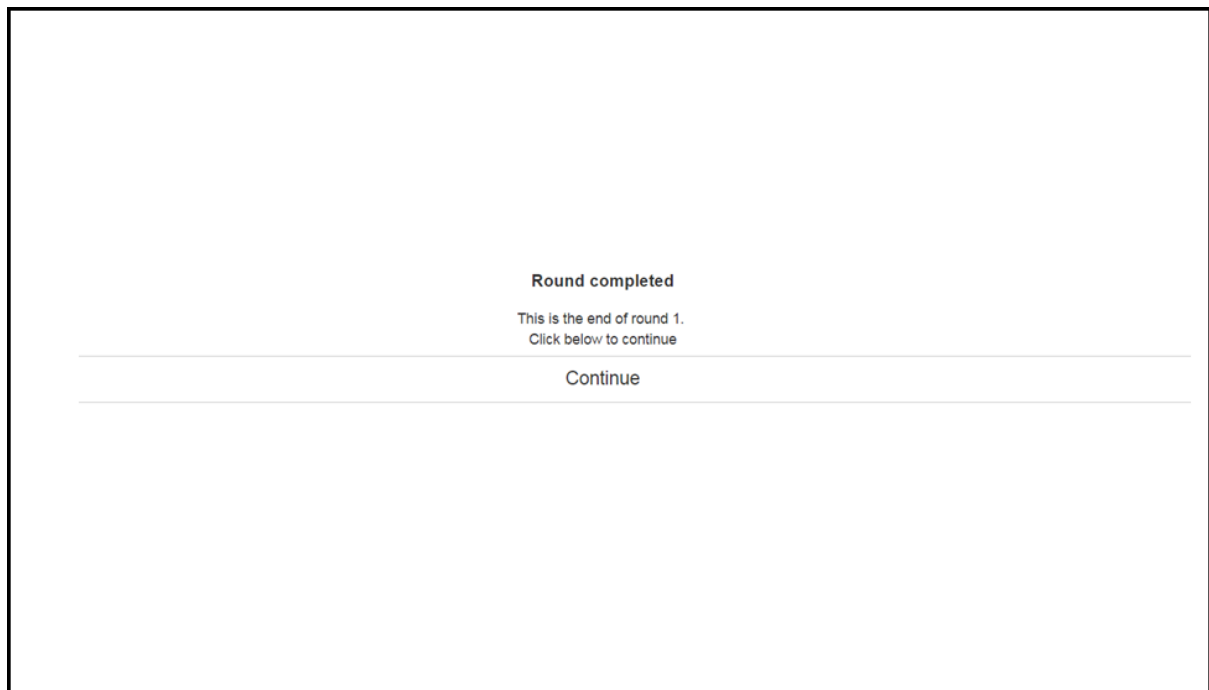


High variance, cluster close to E1 (HC). Two peers showed moderate agreement with the participant, while one peer strongly disagreed.

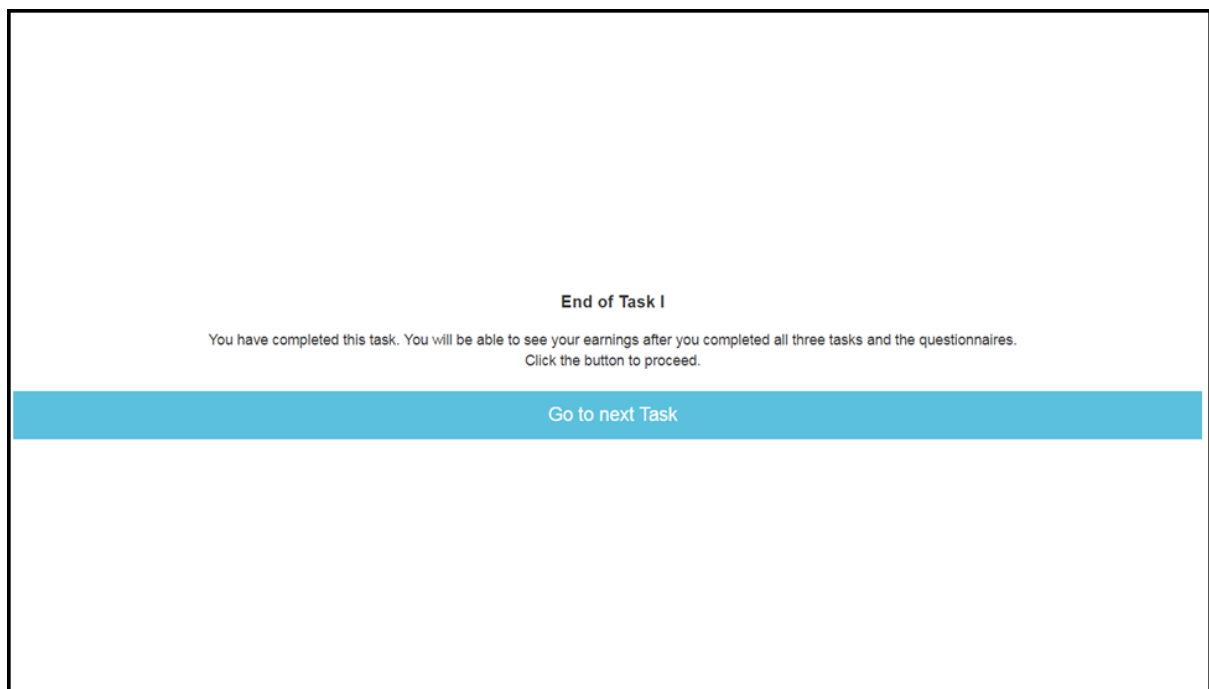


Random condition. Peer estimates were drawn randomly from the prerecorded pool.

After completing 5 rounds for each experimental condition, plus 10 filler rounds (so 30 rounds in total), participants proceeded to the next block.



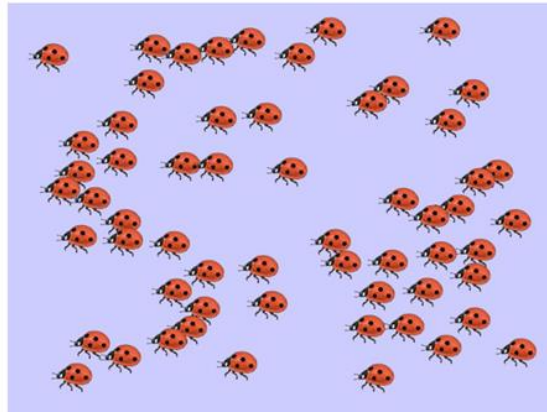
After the 30 rounds of this task were completed, this Block ended, and participants proceeded to Block 2 (referred to in the instructions as 'Task II').



Task II: instructions 1 of 3

This task consists of 5 rounds.

At the beginning of each round, you will observe an image showing a number of animals.
For example:



The image will disappear after 6 seconds, upon which you have to estimate how many animals were displayed.
The more accurate your estimate, the more points you can earn. We explain this later.

Continue

Task II: instructions 2 of 3

Once the image has disappeared, you have to enter your estimate of how many animals were displayed, by using a slider.
This is your estimate for **part A** of a round.

Once you have entered your estimate, **part B** of the round begins.
You can observe the *part A* estimate of another MTurker.

Over 100 MTurkers participated in a previous session in which they completed this task.
(These are different MTurkers from those you could observe in the previous task.)
In each round, you can observe the *part A* estimate of **one** of these previous MTurkers.

The previous MTurkers saw the same image as you. They also saw it for **6 seconds**.
After the image disappeared, they also had to **estimate** how many animals were displayed.
They could also earn a higher bonus if their estimate was more accurate.

You then have to **enter a second estimate**.
You can enter the **same** estimate as in *part A*, or adjust it as you wish.
This is your estimate for **part B** of a round.

*Note: please enter your estimates **within the time limit** on your screen.
If you do not make your estimate and press 'Continue' before the timer reaches zero,
you will be removed from the HIT and we will not be able to pay you!*

Once you have entered your *part B* estimate, the round is over and a new round begins.

Continue

Go back

Check for understanding

To check your understanding of the task, please indicate for each of these statements whether they are correct or incorrect.

In each round of this task you will view an image. You have to estimate how many animals were displayed in it.

correct

incorrect

Once you have entered your estimate, you can observe the estimate of another MTurker who completed this task before. You can then make a second estimate.

correct

incorrect

The more accurate your estimates, the more points you can earn.

correct

incorrect

Continue

Back to instructions

Again, participants had to complete all items correctly before they could proceed to the task.

Round 1

You have answered all control questions correctly. The task starts now.

When you click below, an image will appear showing a number of ants.

After 6 seconds, the image disappears.

You have to estimate how many ants there were, using a slider.

You have to enter your estimate within the time limit.

Click below when you are ready.

Continue

Round 1: part B

The **blue square** shows your part A estimate.
The **red dot** shows the estimate of another MTurker.
Please use the slider to indicate your second estimate.

Your estimate is: **70**



Continue

Remaining time: 00:28

The screen for entering their first estimate was identical to the screen in the previous Block. In this one-peer (control) condition, participants could observe a single peer (here shown in red). The participant's first estimate is shown as a blue square. Participants used the slider to enter their second estimate.

End of Task II

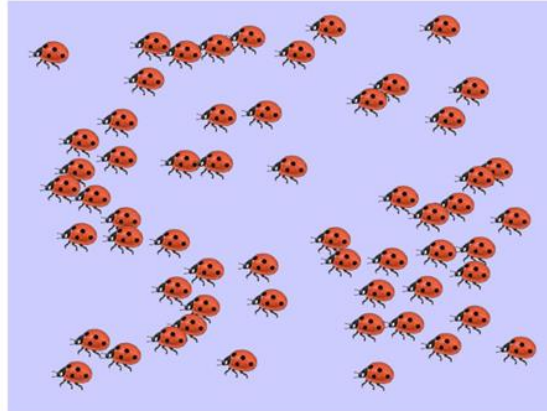
You have completed this task. You will be able to see your earnings after you completed all three tasks and the questionnaires.
Click the button to proceed.

Go to next Task

Task III: instructions 1 of 4

This task consists of 5 rounds.

In each round, you have to make an estimate, based on estimates of other MTurkers.
Specifically, in a HIT we conducted recently, we asked MTurkers to view 5 pictures.
Each picture displayed a number of animals.
For example:



In each round, the image disappeared after 6 seconds, upon which the previous MTurkers had to estimate how many animals were displayed.
The more accurate their estimates, the more points they could earn.

Continue

Task III: instructions 2 of 4

In this task, you have to estimate the actual number of animals *without seeing the picture yourself*.

That is, in this task you will not see the pictures themselves, but you will **only see the estimates** of the MTurkers who participated previously.

Continue

Go back

Task III: instructions 3 of 4

Over 100 MTurkers participated in a previous session in which they completed their task.
(These are different MTurkers from those you could observe in the previous tasks.)
The estimates of four of these previous MTurkers will be displayed as **red dots** on a slider. This is an example:



You will be able to move the **slider handle** to enter your estimate.

*Note: please enter your estimates **within the time limit** on your screen.
If you do not make your estimate and press 'Continue' before the timer reaches zero,
you will be removed from the HIT and we will not be able to pay you!*

Once you have entered your estimate, the round is over and a new round begins.

Continue

Go back

Task III: instructions 4 of 4

Your bonus earnings

The more accurate your estimates, the more points you can earn in this task.
At the end of this HIT, the points you earn are converted into your bonus earnings.
Your bonus for this task is calculated as follows.

Once you have completed this HIT, the computer will randomly select 1 of the 5 rounds.

If you estimated the number of animals **exactly right**, you earn **100 points**.

For each number that you are off, we subtract 5 points.

The number of points you earn cannot become negative.

For example, imagine that the actual number of animals in the image that the previous MTurkers saw was 60.
If your estimation was 53, you were 7 off.
This would mean that we subtract $7 \times 5 = 35$ points. Your earnings for that estimate would be $100 - 35 = 65$ points.

Click 'Continue' if you understood your task.
A brief quiz will follow to check your understanding.

Continue

Go back

Check for understanding

To check your understanding of the task, please indicate for each of these statements whether they are correct or incorrect.

In each round of this task you have to estimate how many animals were shown to previous MTurkers.

correct

incorrect

In each round, you will see the estimates of four MTurkers who all saw the same picture.

correct

incorrect

Like yourself, the previous MTurkers could earn more points the more accurate their estimates were.

correct

incorrect

Continue

Back to instructions

Again, participants had to complete all items correctly before they could proceed to the task.

Round 1

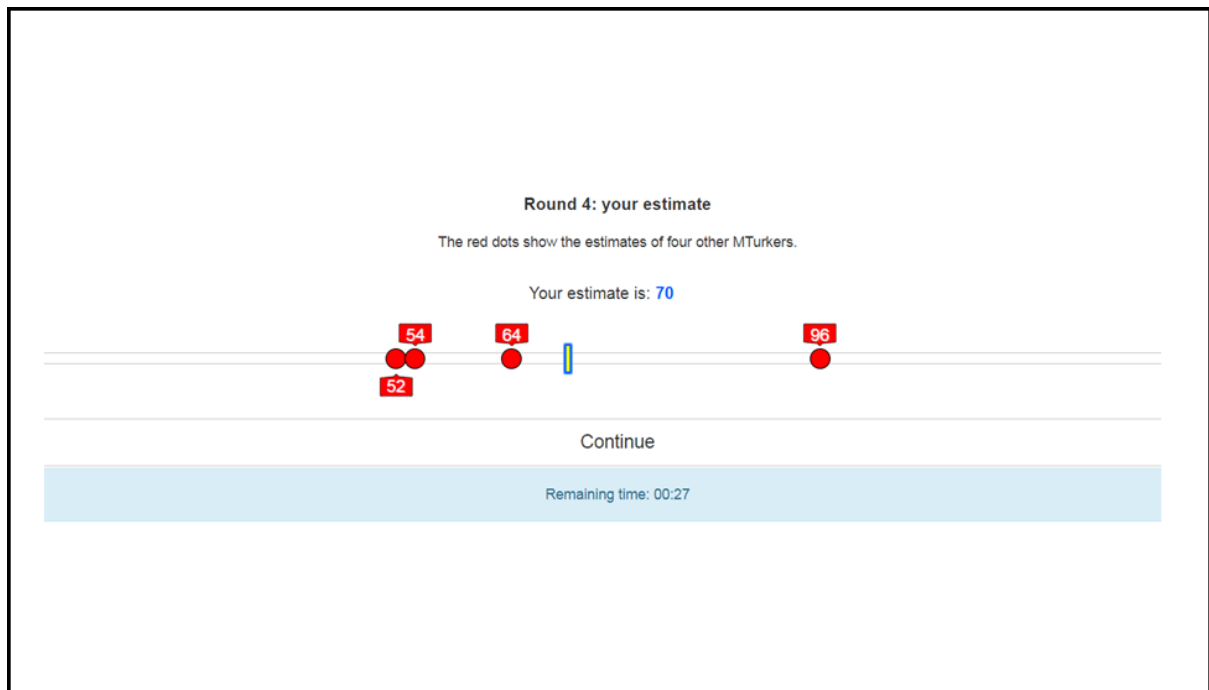
On the next screen, you will see the estimates of four other MTurkers.

All of them saw the same image.

You have answered all control questions correctly! The task starts now.

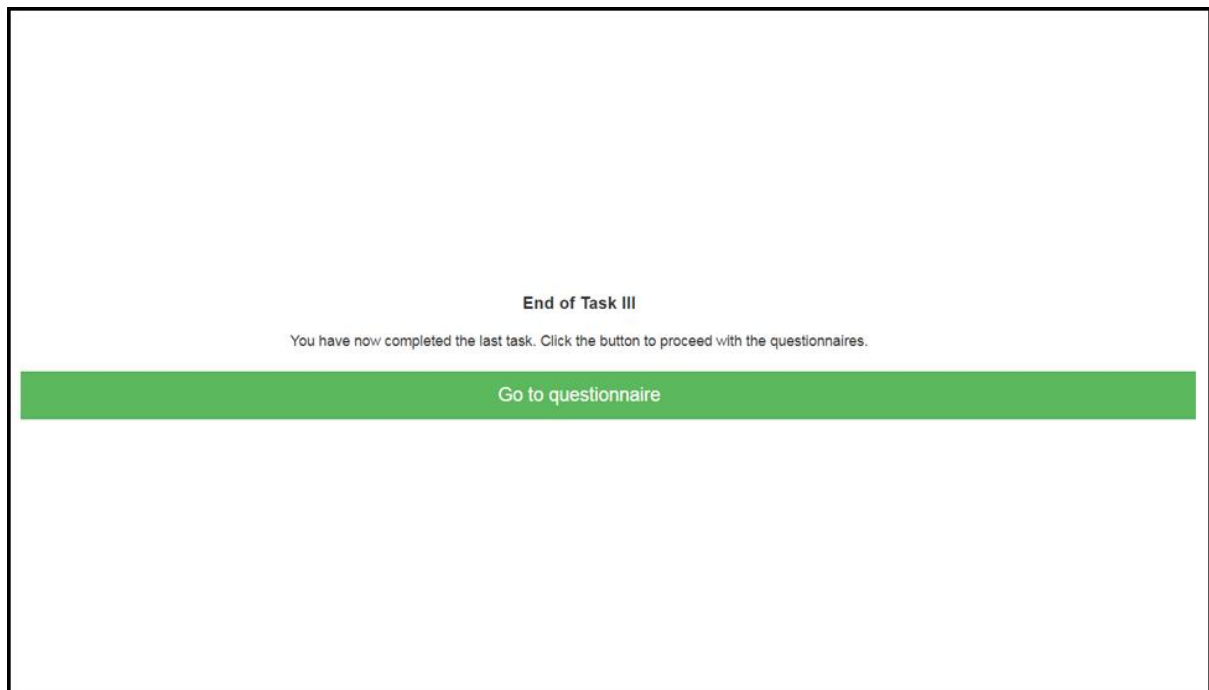
Click below to start the round and estimate the actual number of animals in the image they saw.

Continue

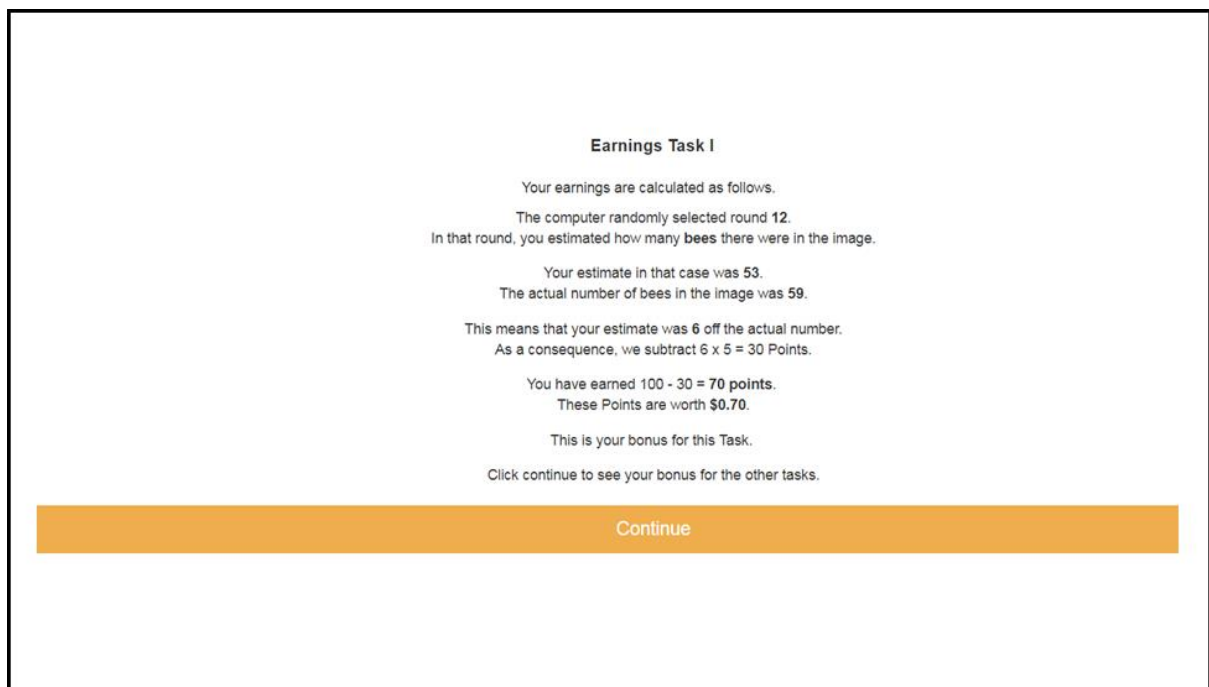


Participants did not observe an image but just observed 4 peer estimates and made their first estimate with the slider. The conditions in this block emulated the conditions in the three-peer conditions (see Methods in the main text for details). This screenshot and the next show two examples.





After completing the third Block, participants proceeded to a questionnaire, in which we measured age and gender (along with some other items not reported here). Finally, participants were shown their payoffs for each of the blocks.



Earnings Task II

Your earnings are calculated as follows.

For Task I, the computer randomly selected round 3.
In that round, you estimated how many **flamingos** there were in the image.

Your estimate in that case was 75.
The actual number of flamingo in the image was 59.

This means that your estimate was 16 off the actual number.
As a consequence, we subtract $16 \times 5 = 80$ Points.

You have earned $100 - 80 = 20$ points.
These Points are worth \$0.20.

This is your bonus for this Task.

Click continue to know your bonus for the other tasks.

Continue

Earnings Task III

Your earnings are calculated as follows.

The computer randomly selected round 2.
In that round, you estimated how many **animals** there were in the image that other MTurkers saw.

Your estimate in that case was 45.
The actual number of bees in the image was 77.

This means that your estimate was 32 off the actual number.
As a consequence, we would have subtracted 32×5 Points.
However, as stated in the instructions, your number of Points cannot become negative, so we subtract 100 Points.

You have earned $100 - 100 = 0$ points.
These Points are worth \$0.00.

This is your bonus for this Task.

Click continue to know your bonus for the other tasks.

Continue

Total earnings

This is the end of this HIT. Your earnings are calculated by adding up your Points from Task I, Task II Task III:

20 (from Task I) *plus* 70 (from Task II) *plus* 0 (from Task III)= **90 Points**.

These Points are worth **\$0.90**. This is your bonus for this HIT.

Note that any bonus you earn will be paid on top of your guaranteed participation fee of **\$4.50**.

To collect your payment for the HIT you just completed, please fill out the below code on MTurk.

1000108

Once you have done that, you can close this window.

Thank you for your participation, and we hope you enjoyed participating in our HIT.